SECTION 2 – ROOM LAYOUTS

TABLE OF CONTENTS

2–1 INTRODUCTION

When laying out a floor plan there are special considerations that must be taken into account due to the magnetic field effect on equipment. The maximum magnetic field in which the upgrade supplied equipment can be located is listed in Table 2–1.

TABLE 2–1
PROXIMITY LIMITS

GAUSS (mT) LIMIT See Notes 1 & 2	EQUIPMENT	
0.5 GAUSS (0.05mT) OR LESS See Note 4	• Nuclear cameras	
1 GAUSS (0.1mT) OR LESS See Note 4	• Positron Emission Tomography scanner • Linear Accelerator • Cyclotrons • Accurate Measuring scale • Image intensifiers • Color TV	• Video display (color, B/W, monochrome) • CT scanner • Ultrasound • Lithotriptor • Electron microscope • Advantage Workstation with CRT Monitor
3 GAUSS (0.3mT) OR LESS See Note 4	• Power transformers • Main electrical distribution transformers • Moving steel equipment such as: – Vehicular traffic – Fork lift trucks – Dumb waiters – Electric transport carts	– Loading dock (truck traffic) – Elevators – Escalators – Helicopters (See Note 3)
1 GAUSS (0.1mT) OR LESS	• Advantage Workstation with CRT Monitor	
10 GAUSS (1mT) OR LESS	• Operator Workspace Host Monitor (Color)	
30 GAUSS (3mT) OR LESS	• System Control Cabinet • Twin Accessory Cabinet	• TSCC equipment • TGWC equipment
50 GAUSS (5mT) OR LESS	• Octane Computer • Operator Workspace Cabinet • LCD Color Monitor for OW (See note 5) • SRF Cabinet • ACGD/PDU Cabinet • Main Disconnect Device	• Magnet Monitor • Telephones • Metal detector for screening • LCD Color Monitor for Advantage Workstation • MNS Cabinet
100 GAUSS (10mT) OR LESS	• Service Tool Magnet Power Supply Cabinet • Service Tool Shim Power Supply Cabinet	• Shield/Cryo Cooler Compressor Cabinet • Pneumatic Patient Alert Control Box
200 GAUSS (20mT) OR LESS	• Penetration Panel • Blower Box	• Magnet Rundown Unit

Note
1. Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable.
2. Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable.
3. Verify operating limits with provider of helicopter service
4. The 0.5 gauss (0.05 mT), 1 gauss (0.1 mT), and 3 gauss (0.3 mT) or less proximity limits apply to ALL magnets. For additional Active Shield magnet proximity limits refer to ***Direction 2294003 Signa 1.5T TwinSpeed Pre–Installation*** **Section 4–13 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS**.
5. If gauss limit is more than indicated for OW with LCD Color Monitor then contact GE to determine impacts on OW equipment.

2–2 TWO MAGNET SITE LAYOUT

Refer to ***Direction 2294003 Signa 1.5T TwinSpeed Pre–Installation*** **Section 2–3 TWO MAGNET SITE LAYOUT** for two magnet site layout information.

2–3 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY

Table 2–2 lists minimum actual clearance opening dimensions for doors and hallways required by new equipment installed as part of the various Signa 1.5T TwinSpeed Upgrades. Installation or replacement of components listed in Table 2–3 must be taken into consideration when determining delivery route hallway and door dimensions. Clearance for maneuvering around corners or turns must also be taken into consideration. Refer to SECTION 8, SHIPPING AND DELIVERY DATA, for component shipping dimensions.

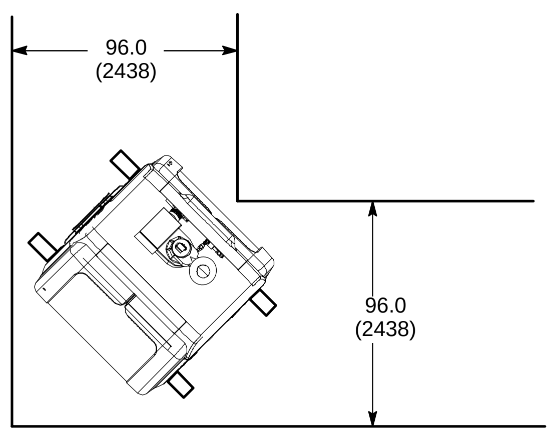
TABLE 2–2
MINIMUM HALLWAY/DOOR DIMENSIONS

COMPONENT	MINIMUM HALLWAY/ DOOR WIDTH* in. (mm)	MINIMUM HALLWAY/ DOOR HEIGHT* in. (mm)	COMMENTS
Operator Workspace Table	32 (813)	80 (2032)	
Indoor Water Cooled TSCC [Model LC18MT/WC]	48 (1219)	80 (2032)	
Indoor Air Cooled TSCC [Model LC 20MT]	50 (1270)	80 (2032)	
Equipment Cabinets	36 (914)	80 (2032)	Includes Indoor Water Cooled TGWC [Model LTL 10] or Indoor Air Cooled TGWC [Model LC 10M]
Cryogen delivery route and Storage Room	43 (1092)	80 (2032)	Width requirements due to size of 500 liter dewars. Width and height requirements vary dependent on the dewars used. Check with cryogen supplier.
LCC Magnet	Refer to Note 1	Refer to Note 2	Refer to Table 2–3 for uncrated magnet dimensions.
RF Room Door	Refer to Section 7, RF SHIELDED ROOM	Refer to Section 7, RF SHIELDED ROOM	

Note * Minimum hallway and door dimensions are actual clearance openings. Width and height of rigging equipment is not included in above dimension.

1 Minimum width depends on access route to removable panels of RF shielded room wall. For straight path (i.e. no bends or turns) recommended to allow 6 in. (153 mm) on both sides of magnet. Appropriate calculations must be performed if turns exist along proposed magnet delivery route. Illustration 2–1 shows dimensions for 90° turn.

2 Final dimension is dependent on rigger equipment used, refer to ***Direction 2294003 Signa 1.5T TwinSpeed Pre–Installation*** SECTION 8, SHIPPING AND DELIVERY DATA.



NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

MAGNET MINIMUM DOOR/HALLWAY DIMENSIONS 90° TURN
ILLUSTRATION 2–1

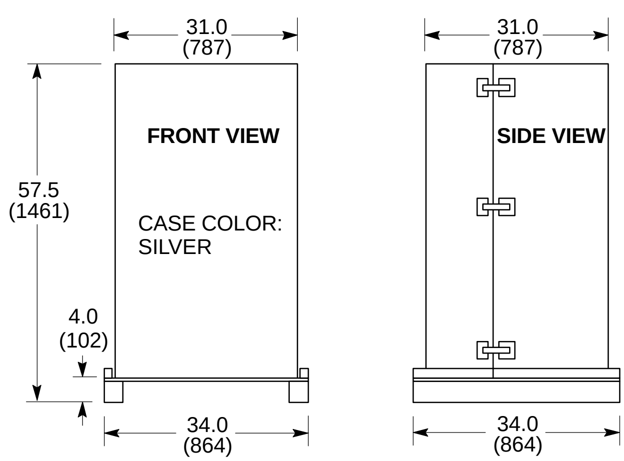
2–3 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY (Continued)

TABLE 2–3
UPGRADE COMPONENT DIMENSIONS FOR INSTALLATION/REPLACEMENT

COMPONENT	APPROXIMATE WEIGHT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	COMMENTS
LCC Magnet (uncrated)	Refer to comments.	Refer to comments.	Refer to ***Direction 2294003 Signa 1.5T TwinSpeed Pre–Installation*** Section 8, SHIPPING AND DELIVERY DATA, for dimensions, illustrations and weights.
Split Bridge (LCC Magnet Enclosure)	40 (18)	21.5 x 77.3, x 7 (546 x 1969 x 18)	Room dimensions in front of LCC Magnet MUST allow for bridge installation/servicing and Gradient Coil Replacement, See Note 3.
Replacement Gradient (TRM) Coil Assembly on a Shipping Cradle/Cart	See Note 1	35.5 x 96.09 x 55.88 (902 x 2441 x 1420) See Note 2 & 3	Shipping/installation cart is used to install replacement coil assembly only. Refer to Section 2–3–1, Gradient Coil Assembly, Cradle, & Cart Requirements.
Gradient (TRM) Coil Replacement Tool Kit Crate for LCC Magnets	750 (340)	30 x 86 x 28 (762 x 2184 x 711)	See Note 3.
Upgrade or Replacement RF Body Coil	245 (111)	31 x 31 x 57.5 (787 x 787 x 1461)	Replacement coil is shipped in a protective case. Weight & dimensions are for coil & case, see Illustration 2–2.
Enclosure Covers & End Bells	See Comments	See Comments	Refer to Section 2–9 COMPONENT DIMENSIONS, for details of individual assemblies and applicable catalogs.
Rear Pedestal	30 (13.6)	17 x 43 x 33 (432 x 1092 x 838)	

Note

1. The replacement TRM Gradient Coil Assembly weight is approximately 2850 lbs (1293 kg), the shipping cradle is 125 lbs (57 kg), and the Gradient Coil Assembly shipping/installation cart weighs 855 lbs (389 kg). Therefore total shipping weight is 3830 lbs (1738 kg). The coil assembly outside diameter x length dimensions are 35.0 x 74.6 in. (888 x 1895 mm).
2. Gradient (TRM) Coil Assembly and shipping cart dimensions are with cart in lowest position. Cart can be adjusted to maximum height of 61.88 in. (1572 mm).
3. LCC Magnets MUST USE GE Service Tool Gradient Coil Replacement Kit for replacing the Gradient Coil Assembly. The Gradient Coil Replacement Tool Kit is shipped in a wooden crate on casters. Utilization of Gradient Coil Replacement Tool requires 107.75 in. (2737 mm) clear space in front of magnet.



NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 245 lbs (111 kg)
  WEIGHT AND DIMENSION INCLUDE COIL, CASE, AND SHIPPING PALLET.
- RF COIL CASE HAS WHEELS AND HANDLES FOR EASE MOVING AFTER REMOVAL FROM PALLET.

RF COIL CASE – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE FOR SIGNA 1.5T TWINSPEED UPGRADE
ILLUSTRATION 2–2

2–3 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY(Continued)

2–3–1 Gradient Coil Assembly, Cradle, & Cart Requirements

Systems With LCC Magnet Currently Installed

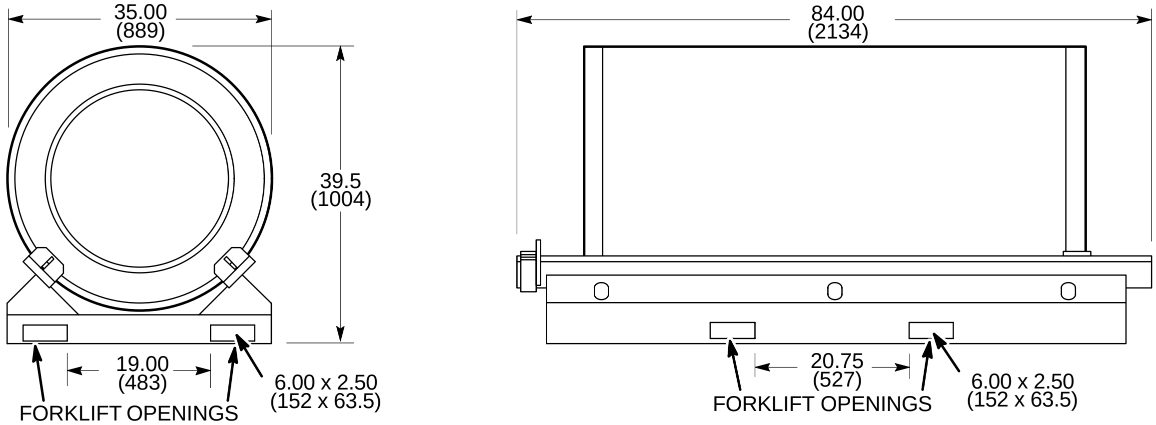
Systems with an LCC Magnet currently installed will require the Body Coil Insertion Tool to upgrade from BRM or CRM to the new TRM coil. The Body Coil Insertion Tool requires the TRM coil to be installed on the aluminum cradle (see Illustration 2–1). and be mated onto the aluminum coil cart (see Illustration 2–2). The new TRM Coil will be shipped on a wooden disposable cradle. The TRM Coil **must** be transferred to an empty aluminum cradle and cart provided GE Service via the Tools Depot. GE Service provided coil lifting bracket will be necessary for transferring the coil and a customer rigger provided forklift or crane/hoist rated for 4,000 lbs (1814 kg). Refer to **Section 10–3 RIGGER/CUSTOMER SUPPLIED EQUIPMENT** for additional rigger/customer supplied equipment. Refer to **Section 10–5 INSTALLATION EQUIPMENT** for GE Service provided equipment.

Systems Installing LCC Magnet Upgrade Concurrent With TwinSpeed Upgrade

Systems installing an LCC Magnet Upgrade at the same time as the TwinSpeed Upgrade Systems will receive (M3060TA) LCC Magnet with TwinSpeed Enclosure with the TRM Assembly (M3085TA) installed inside the magnet bore and enclosure.

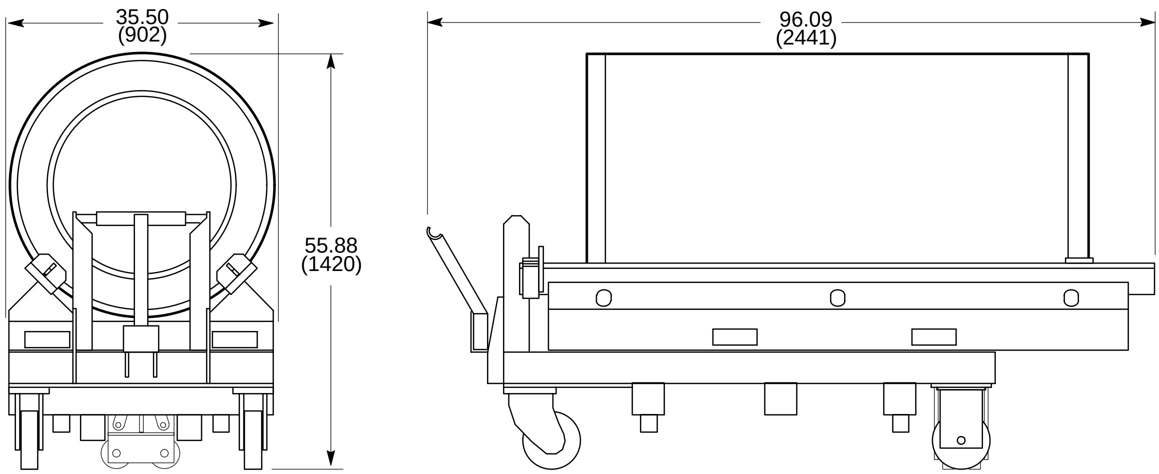
2–3–1 Gradient Coil Assembly, Cradle, & Cart Requirements (Continued)

NOTE:
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT WITH TRM 2975 lbs (1349 kg)



GRADIENT (TRM) COIL ASSEMBLY & ALUMINUM CRADLE
ILLUSTRATION 2–1

NOTE:
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT WITH TRM 3830 lbs (1737 kg)



SHORT ROOM REPLACEMENT GRADIENT (TRM) COIL ASSEMBLY & CRADLE ON CART
ILLUSTRATION 2–2

2–4 CABLING CONSIDERATIONS

Cable runs in the Magnet Room as well as throughout the system must be in accordance with local and national codes. Cabinets cable access areas are shown on illustrations in Section 2–9, COMPONENT DIMENSIONS. Signa 1.5T TwinSpeed upgrade results in several changes to the cabling and hoses routed through the Rear Pedestal cable access areas of the Magnet Enclosure. These changes include the

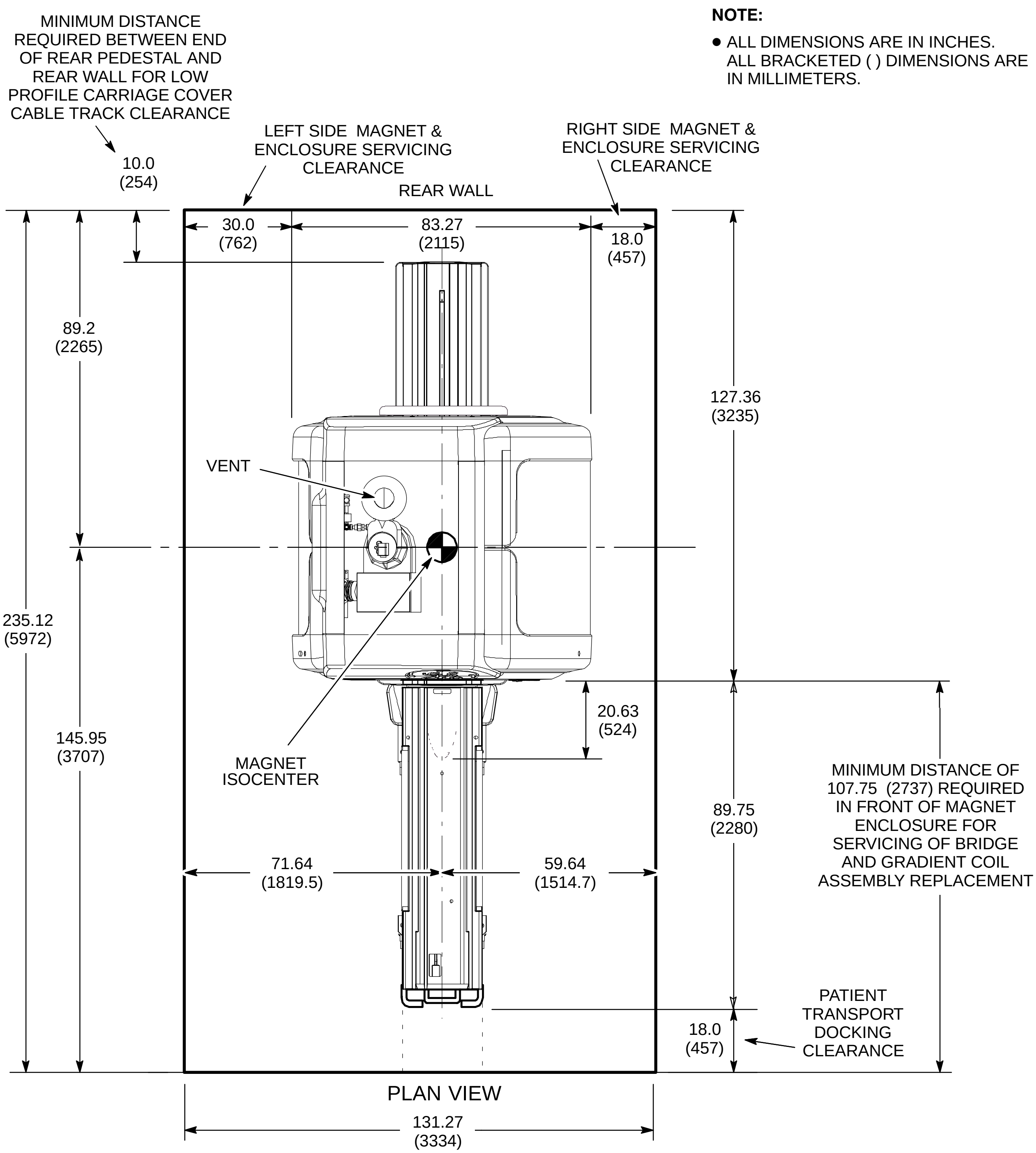
- elimination of 1 Blower Box hose (6.5 inch (165 mm) OD hose)
- installation of 3 additional Gradient cables
- addition of a vacuum hose and a vacuum sensor cable for the Quiet Technology Enclosure
- 2 existing water hoses for Body Coil are replaced with 2 new larger water hoses.

Refer to Section 6 – INTERCONNECT DATA for cabling and hoses details.

The remainder of the Magnet Enclosure system interconnect cables are also still routed through the Rear Pedestal cable access. The existing cable access area utilized to site cable routing should be sufficient for the upgrade system cabling. See Illustration 2–5 for Magnet cable access areas and refer to Section 6 – INTERCONNECT DATA for cables and their dimensions information.

If an LCC Magnet upgrade is being performed with the system TwinSpeed upgrade there will be additional system interconnects changes for the new magnet. Refer to Section 6 – INTERCONNECT DATA for additional information and details.

2–4 CABLING CONSIDERATIONS (Continued)

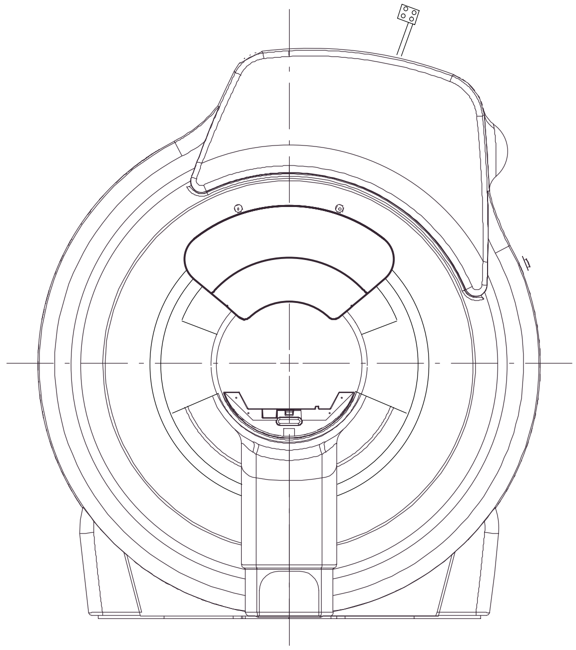


SIGNA TWIN*SPEED* 1.5T LCC (ACTIVE SHIELD) MAGNET (MINIMUM SERVICE AREA)
ILLUSTRATION 2–3

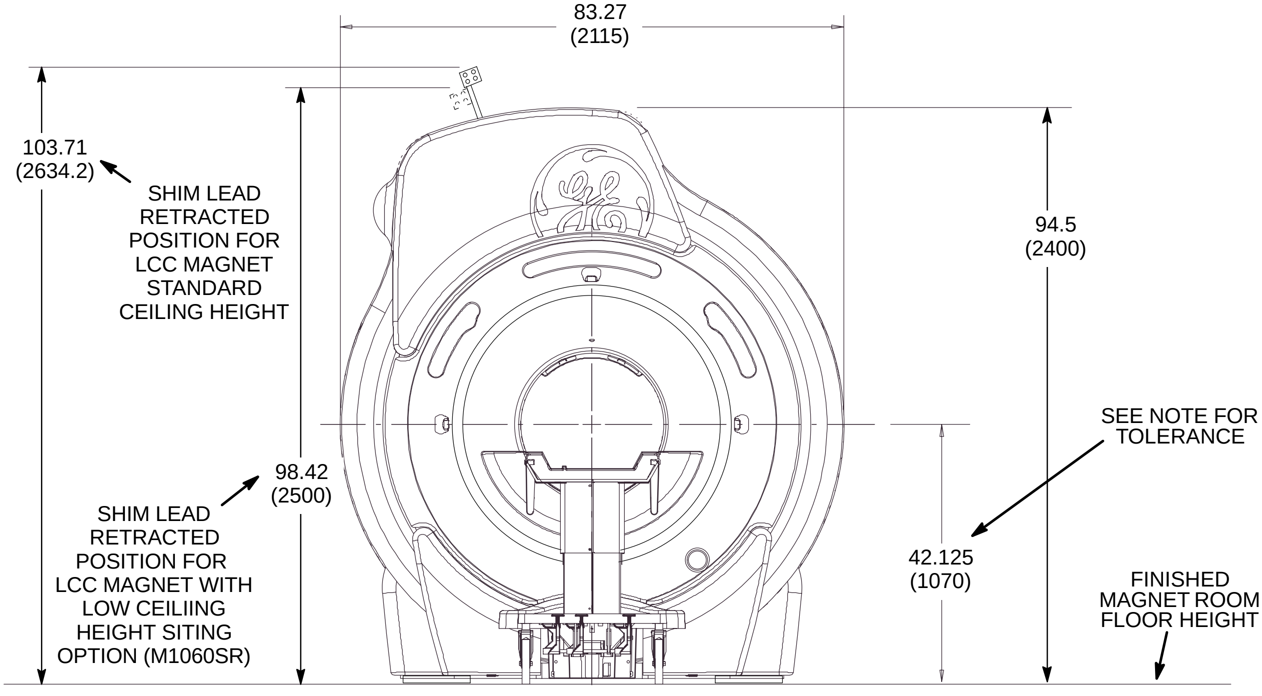
2–4 CABLING CONSIDERATIONS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- FINISHED FLOOR TO MAGNET CENTER LINE HEIGHT DIMENSION MUST BE 42.125±0.25 inches (1070±6.35 mm) TO ALLOW PATIENT TABLE TO PROPERLY DOCK TO THE ENCLOSURE.



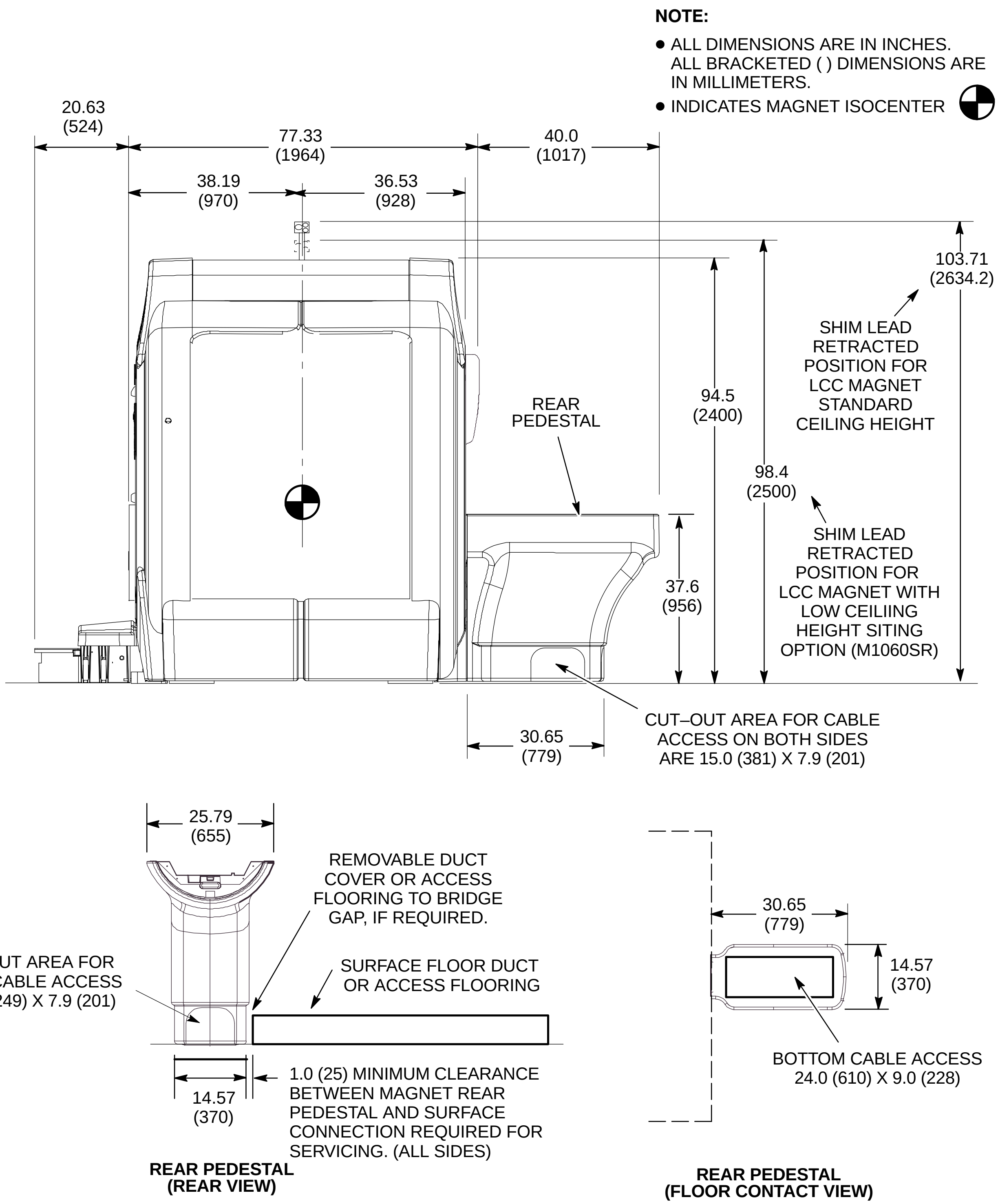
REAR VIEW



FRONT VIEW

SIGNA TWIN*SPEED* 1.5T LCC (ACTIVE SHIELD) MAGNET ENCLOSURE (FRONT AND REAR VIEWS)
ILLUSTRATION 2–4

2–4 CABLING CONSIDERATIONS (Continued)



SIGNA TWIN*SPEED* 1.5T LCC (ACTIVE SHIELD) MAGNET CABLE ACCESS
ILLUSTRATION 2–5

2–5 System Cooling Equipment

The Signa TwinSpeed system requires water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the TRM Gradient Coil located in the Magnet bore. The Signa TwinSpeed system requires a different capacity water cooling unit for the TRM Gradient Coil located in the Magnet bore from the units utilized in prior system configurations. If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then the Shield/Cryo Cooler Compressor will be required. Refer to **Section 1 – SYSTEM CONFIGURATION** flowcharts for listing of catalogs dependent on system site configuration.

- Twin System Cooling Cabinet (TSCC) which provides water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil located in the Magnet bore. Refer to Section 2–5–1 TSCC Siting Considerations for details.
- Twin Gradient Water Cooler (TGWC) which provides water cooling for ONLY the Gradient Coil located in the Magnet bore. When a TGWC is selected then the customer site must still provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. Refer to Section 2–5–2 TGWC Siting Considerations for details.

Note

Transportable/Relocatable Site Signa 1.5T TwinSpeed upgrade will replace the existing van manufacturer provided water cooling unit with the Outdoor Air Cooled Twin System Cooling Cabinet (TSCC) unit.

2–5–1 TSCC Configurations Siting Considerations

CAUTION

Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore MUST be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival. Therefore the Twin System Cooling Cabinet (TSCC) and Main Disconnect Panel (MDP) must be installed and operational prior to magnet arrival.

Note

Consult local/national code for interconnects separation requirements (i.e. signal, power, water, etc.).

The TSCC is available in three fixed site configuration options:

- Outdoor Air Cooled TSCC [Model LC 20MT/O] has the TSCC unit located outside the building and a Remote Control Panel (RCP) located in the MR system Equipment Room; see Illustration 2–10 for Outdoor Air Cooled TSCC and Illustration 2–12 for the Remote Control Panel.
- Indoor Water Cooled TSCC [MODEL LC 18MT/WC]; see Illustration 2–13 and refer to Section 4–4–3 Water Cooled TSCC Requirements for water cooling requirements.
- Indoor Air Cooled TSCC [MODEL LC 20MT]; see Illustration 2–15.

Note

Customer provided temporary backup water cooling is recommended for the Shield/Cryo Cooler Compressor Cabinet. Refer to Section 4–4–2 Shield/Cryo Cooler Compressor Temporary Backup Water Cooling Recommendation.

2–5–1 TSCC Configurations Siting Considerations (Continued)

TSCC & System Interconnects/Separation Limitations

Location of the Outdoor TSCC must meet the following limitations for the water lines (customer provided copper line and E&W/GE provided flexible tubing combined).

- Outdoor TSCC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 200 ft (61 m) water line length.
- Outdoor TSCC and the Shield/Cryo Cooler Compressor Cabinet must not be separated by a distance greater than 160 ft (48.8 m) water line length.
- The Outdoor TSCC vertical separation from the Gradient Coil located inside the magnet and the Shield/Cryo Cooler Compressor must not exceed 100 ft (30.5 m) above or below the Gradient Coil and Shield/Cryo Cooler Compressor.
- The Outdoor TSCC and the RCP must not be separated by a distance greater than 100 ft (30.5 m) total length.

Location of the Indoor Water Cooled TSCC or Indoor Air Cooled TSCC must meet the following limitations for the water lines (E&W/GE provided flexible tubing).

- Indoor Water Cooled TSCC or Indoor Air Cooled TSCC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) water line length.
- Indoor Water Cooled TSCC or Indoor Air Cooled TSCC and the Shield/Cryo Cooler Compressor Cabinet must not be separated by a distance greater than 60 ft (18.3 m) water line length.
- The Indoor Water Cooled TSCC or Indoor Air Cooled TSCC vertical separation from the Gradient Coil located inside the magnet and the Shield/Cryo Cooler Compressor must not exceed 15 ft (4.6 m) above or below the Gradient Coil and Shield/Cryo Cooler Compressor.

The TSCC is powered from the system Main Disconnect Panel (MDP) via customer supplied wiring. Refer to **Section 6, INTERCONNECT DATA**, for additional interconnects information and requirements.

2–5–1 TSCC Configurations Siting Considerations (Continued)

Responsibility For Installation Tasks For TSCC Equipment

The TSCC subsystem equipment installation requires specific tasks to be performed by the Customer Contractor, GE Service, and Ellis & Watts Service. Table 2–4 lists the responsibility for the specific tasks. Refer to vendor manual for additional information concerning tasks.

TABLE 2–4
TSCC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED

APPLICABLE CONFIG				RESPONSIBILITY		
OUTDOOR (LC20MT/O)	**INDOOR WATER COOLED (LC18MT/W)**	**INDOOR AIR COOLED (LC20MT)**	**INSTALLATION TASKS FOR TSCC EQUIPMENT** See Notes 1, 2, & 3	**CUSTOMER CONTRACTOR**	**GE SERVICE**	**E&W SERVICE PROVIDER**
			WHEN TSCC IS AT SITE & PRIOR TO MAGNET DELIVERY			
X	X	X	Unload TSCC from truck.	X		
X			Move TSCC to site outdoor location and mount. (Use seismic tie down brackets where required by law)	X		
X			Install Remote Control Panel in Equipment Room	X		
	X		Move TSCC to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
		X	Move TSCC to Equipment Room and mount. (Use seismic tie down brackets & vibration isolators where required by law)	X		
X	X		Install customer water supply and return lines to TSCC	X		
X	X	X	Connect customer supplied power wiring from MDP to TSCC.	X		
X			Install power and control cable between TSCC & RCP	X		
			WHEN MAGNET IS DELIVERED/INSTALLED			
X			Install water cooling lines from customer supplied/installed ball values & hose barbs on copper lines from TSCC to 1) Gradient Coil and 2) Shield/Cryo Cooler Compressor. Refer to **Section 6 – INTERCONNECT DATA** for details of interconnects.		X	
	X	X	Install water cooling lines from TSCC to 1) Gradient Coil and 2) Shield/Cryo Cooler Compressor. Refer to **Section 6 – INTERCONNECT DATA** for details of interconnects.		X	
X	X	X	Fill TSCC with Propylene Glycol		X	
X	X	X	See **APPENDIX B Prerequisites for Scheduled Performance Verification of Chillers For Signa Twin*Speed*** and review for completeness. Send email request for initial performance verification to service@elliswatts.com. Email to include: System ID, Magnet Monitor System ID, Model and Serial Number of the SCC unit, and Customer location.		X	
X	X	X	Verify installation and operation of TSCC per E&W Chiller Startup Checklist			X

Note
1 Refer to Ellis & Watts *Technical Publication 470 Pre installation/Installation/Operating LC 20MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Outdoor TSCC information and details.

2 Refer to Ellis & Watts *Technical Publication 471 Pre installation/Installation/Operating LC 18MT/WC Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Indoor Water Cooled TSCC information and details.

3 Refer to Ellis & Watts *Technical Publication 472 Pre installation/Installation/Operating LC 20MT Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Indoor Air Cooled TSCC information and details.

2–5–2 TGWC Configurations Siting Considerations

CAUTION

When a TGWC is selected for the system then the customer site must still provide cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet; refer to Section 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC

Note
Consult local/national code for interconnects separation requirements (i.e. signal, power, water, etc.).

The TGWC is available in three fixed site configuration options:

- Outdoor Air Cooled TGWC [MODEL LC 10M/O] has the TGWC unit located outside the building and a Remote Control Panel (RCP) is located in the MR system Equipment Room; see Illustration 2–17 for Outdoor Air Cooled TGWC and Illustration 2–19 for the RCP.
- Indoor Water Cooled TGWC [MODEL LTL 10] has the TGWC unit and a Remote Control Panel both located in the MR system Equipment Room; see Illustration 2–20 and refer to **Section 4–4–5 Water Cooled TGWC Requirements** for water cooling requirements.
- Indoor Air Cooled TGWC [MODEL LC 10M]; see Illustration 2–22.

TGWC & System Interconnects/Separation Limitations

Location of the Outdoor TGWC must meet the following limitations for the water lines (customer provided copper line and GE provided flexible tubing combined).

- Outdoor TGWC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 200 ft (61 m) water line length.
- The Outdoor TGWC vertical separation from the Gradient Coil located inside the magnet must not exceed 100 ft (30.5 m) above or below the Gradient Coil.
- The Outdoor TGWC and the RCP must not be separated by a distance greater than 100 ft (30.5 m) total length.

Location of the Indoor Water Cooled TGWC or Indoor Air Cooled TGWC must meet the following limitations for the water lines (GE provided flexible tubing).

- Indoor Water Cooled TGWC or Indoor Air Cooled TGWC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) water line length.
- The Indoor Water Cooled TGWC or Indoor Air Cooled TGWC vertical separation from the Gradient Coil located inside the magnet must not exceed 15 ft (4.6 m) above or below the Gradient Coil.

The Outdoor TGWC and Indoor Air Cooled TGWC are powered from the Main Disconnect Panel (MDP) via customer supplied wiring. The Indoor Water Cooled TGWC is powered from the MR system PDU. Refer to Section 6, INTERCONNECT DATA, for additional interconnects information and requirements.

2–5–2 TGWC Configurations Siting Considerations (Continued)

Responsibility For Installation Tasks For TGWC Equipment

The TGWC subsystem equipment installation requires specific tasks to be performed by the Customer Contractor, GE Service, and Ellis & Watts Service. Table 2–5 lists the responsibility for the specific tasks. Refer to vendor manual for additional information concerning tasks.

TABLE 2–5
TGWC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED

APPLICABLE CONFIG				RESPONSIBILITY		
OUTDOOR (LC10MT/O)	INDOOR WATER COOLED (LC10)	INDOOR AIR COOLED (LC10MT)	**INSTALLATION TASKS FOR TGWC EQUIPMENT** See Notes 1, 2, & 3	CUSTOMER CONTRACTOR	GE SERVICE	E&W SERVICE PROVIDER
			WHEN TGWC IS AT SITE & PRIOR TO MAGNET DELIVERY			
X	X	X	Unload TGWC from truck.	X		
X			Move TGWC to site outdoor location and mount. (Use seismic tie down brackets where required by law)	X		
X			Install Remote Control Panel in Equipment Room	X		
	X		Move TGWC and Remote Control Panel to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
		X	Move TGWC to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
X	X		Install customer water supply and return lines to TGWC	X		
X		X	Connect customer supplied power wiring from MDP to TGWC.	X		
X	X		Install power and control cable between TGWC & RCP	X		
			WHEN MAGNET IS DELIVERED/INSTALLED			
X			Install water cooling lines from customer supplied/installed ball values & hose barbs on copper lines from TGWC to Gradient Coil. Refer to **Section 6 – INTERCONNECT DATA** for details of interconnects.		X	
	X	X	Install water cooling lines from TGWC to Gradient Coil. Refer to **Section 6 – INTERCONNECT DATA** for details of interconnects.		X	
X		X	Fill TGWC with Propylene Glycol		X	
X		X	Verify installation and operation per E&W Chiller Startup Check List			X
			(Continued)			

2–5–2 TGWC Configurations Siting Considerations (Continued)

TABLE 2–5 (Continued)
TGWC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED

APPLICABLE CONFIG				RESPONSIBILITY		
OUTDOOR (LC10MT/O)	**INDOOR WATER COOLED (LC10)**	**INDOOR AIR COOLED (LC10MT)**	**INSTALLATION TASKS FOR TGWC EQUIPMENT** See Notes 1, 2, & 3	**CUSTOMER CONTRACTOR**	**GE SERVICE**	**E&W SERVICE PROVIDER**
			WHEN MR PDU IS DELIVERED/INSTALLED			
	X		Connect E&W supplied power cable from PDU to TGWC.	X		
	X		Fill TGWC with Propylene Glycol		X	
X	X	X	See **APPENDIX B Prerequisites for Scheduled Performance Verification of Chillers For Signa Twin*Speed*** and review for completeness. Send email request for initial performance verification to service@elliswatts.com. Email to include: System ID, Magnet Monitor System ID, Model and Serial Number of the SCC unit, and Customer location.		X	

Note
1. Refer to Ellis & Watts *Technical Publication 474 Pre installation/Installation/Operating LC 10MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Outdoor TGWC information and details.
2. Refer to Ellis & Watts *Technical Publication 469 Pre installation/Installation/Operating LTL 10 Chilled Water System for GE Twin Magnet MRI Equipment* for additional Indoor Water Cooled TGWC information and details.
3. Refer to Ellis & Watts *Technical Publication 473 Pre installation/Installation/Operating LC 10MT Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Indoor Air Cooled TGWC information and details.

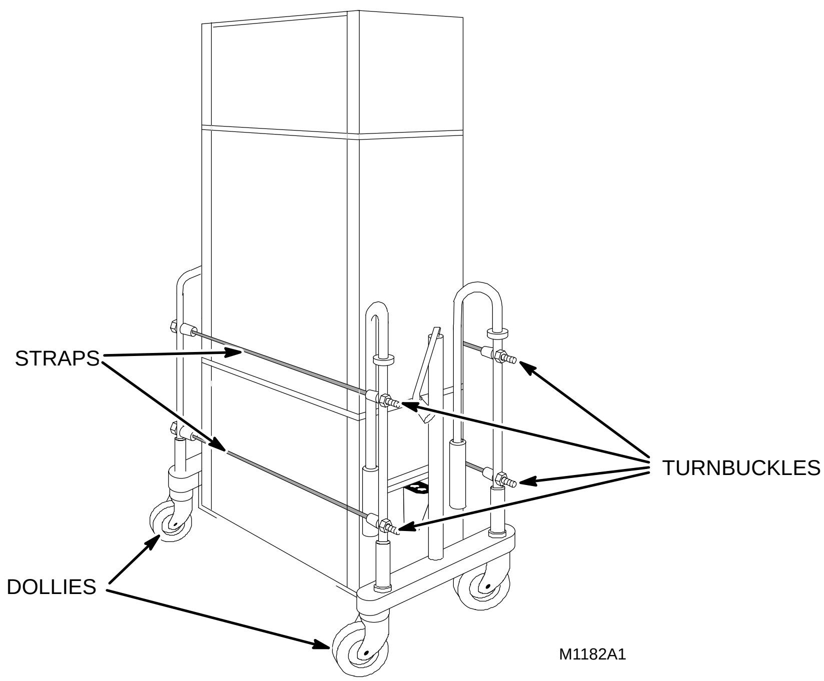
2–6 SPECIAL SITING CONSIDERATIONS

The following system components have special siting concerns which need to be considered.

2–6–1 Power Distribution Unit (PDU)

Sites that are upgrading to ACGD require the Full–size PDU, Compact PDU, or RF/PDU Cabinet be de–installed. The Compact PDU and RF/PDU Cabinet have built–in casters for ease of movement. The Full–size PDU does not have casters and weighs approximately 1000 lbs (455 kg). Sites with a Full–size (double bay) PDU must plan for removal of the PDU.

- **Sites located outside the USA**:
  The Full−size PDU will have to be removed by riggers or other locally supplied equipment moving personnel/equipment.
- **Sites located in the USA:**
  Can contact Techtrans at telephone 800−851−9929 to order a “MR PDU Double Bay Cabinet Dolly Set”. The PDU can be moved with two shipping dollies placed one on each side of the PDU and clamped in place using straps with turnbuckles as shown in Illustration 2–6.



FULL–SIZE PDU SHIPPING DOLLIES
ILLUSTRATION 2–6

2–6–2 Blower Box (MG6)

Systems with the LCC Magnet with a WideOpen Enclosure installed: The Signa 1.5T TwinSpeed upgrade will modify the existing system Blower Box (MG6) mounted within the RF Shielded Room. The 6.5 inch (165.1 mm) OD flexible vinyl air ducting to the body coil is eliminated since the TRM Body Coil is only water cooled. The Blower Box will still be connected to the the Patient Comfort Module of the Quiet Technology Magnet Enclosure with the existing 4.5 inch (114.3 mm) OD flexible vinyl air ducting.

Systems without a WideOpen Enclosure currently installed: The Signa 1.5T TwinSpeed upgrade will add a new Blower Box (MG6) to be mounted within the RF Shielded Room. The new Blower Box will be connected to the the Patient Comfort Module of the Quiet Technology Magnet Enclosure with a 4.5 inch (114.3 mm) OD flexible vinyl air ducting. **The Blower Box contains magnetic material and must be securely mounted to the floor of the Magnet Room or a support shelf on the Magnet Room wall or ceiling with support provided under the box. RF Shield integrity must be maintained for mounting the Blower Box within the Magnet Room.** Refer to Illustrations 2–33 for the new Blower Box and mounting dimensions.

2–6–3 Advantage Workstation Additional Hardware Features (Optional)

The VP Workstation offering is replaced by utilizing an Advantage Workstation with cardiac and neurovascular software offerings. Refer to Direction *2261309–100 Advantage Workstation Ultra60 Pre–Installation* for more information.

The customer must determine the desired location for the AW relative to the imaging system. The following factors should be considered in determining AW location.

- If the customer is purchasing RTIP (neurovascular) option, then the customer may want to observe the experiment real time while operating the Signa system versus reviewing the data at a later time. If the former, then consideration should be made to co–locate the AW near the system Operator Workspace (OW).

Note

LCD monitor is recommended for the AW to be located near the Signa system Operator Workspace due to gauss field proximity limits.

- If the customer is going to perform cardiac studies only, then co–location is not necessary at this time. However, co–location of the AW may be recommended for future applications.

2–6–4 Spectroscopy (Optional)

Signa 1.5T TwinSpeed requires the ACGD/PDU Cabinet. Therefore the spectroscopy option hardware configuration must be compatible with the ACGD system cabinet. Multi–Nuclear Spectroscopy (MNS) option For Systems With ACGD/PDU Cabinet provides the Spectroscopy Acquisition Option which consists of the MNS Cabinet which has mounting space for Broadband RF Amplifier and several module kits to modify the SRF Cabinet, Penetration Panel in the Magnet Room, and system cabling. Refer to Section 6–7, SPECTROSCOPY INTERCONNECTIONS – 1.5T SYSTEM OPTION, cable length information and other cable data.

2–6–5 Magnet Monitor

Refer to ***Direction 2294003 Signa 1.5T TwinSpeed Pre–Installation*** **Section 2–8–7 Magnet Monitor** for Signa 1.5T TwinSpeed LCC Magnet Magnet Monitor and Magnet Monitor UPS option information.

2–6–6 System Monitoring & Support Connectivity Requirements

One of the system monitoring and support connectivity configurations listed in Table 2–6 must be provided for system installation and serviceability purposes. The network connection (if selected configuration) and telephone lines are to be provided and paid for by the customer.

TABLE 2–6
SYSTEM MONITORING & SUPPORT CONNECTIVITY REQUIREMENTS

CONFIGURATION	PHONE LINE	USE/LOCATION
Network connection & telephone lines	One voice–grade telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	Network connection with a static IP address	Access located near the Operator Workspace (OW) in the Control Room.
	One line must be a dedicated direct–distance–dialing voice–grade line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring.
Multiple telephone lines	One voice–grade telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	One line must be a dedicated direct–distance–dialing voice–grade line (data line)	Access located near the Operator Workspace (OW) in the Control Room. **See Note 1 & 2**
	One line must be a dedicated direct–distance–dialing voice–grade line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring. **See Note 1 & 2**
Note 1 A dedicated direct–distance–dialing voice–grade telephone line can be shared for Operator Workspace (OW) and Magnet Monitor (MSM) requirement through the use of a multiplexer box. The following multiplexer boxes are available for customer purchase. 46–328475P1 4 Line Phone Multiplexer box; 115 VAC input power 46–328475P3 4 Line Phone Multiplexer box; 220 VAC input power If the customer chooses not to purchase the multiplexer box then the customer must provide an additional line for each requirement as stated in this table. 2 If a Multiplexer Box is used then the Magnet Monitor **MUST** be Channel 1 to allow for call out after a power outage.		

2–7 ARCHITECTURAL REMINDERS

1. The operator seated at the Operator Workspace must have an unobstructed view of the patient on the transport table when table is docked to the magnet.
2. It is recommended that the Magnet Room viewing window be of fine mesh screening material (as opposed to a “honeycomb–type pattern”) for better visibility of the patient from the Operator Workspace.
3. Customer provided and paid for telephone lines must be supplied for system installation and serviceability purposes per **Section 2–6–6 System Monitoring & Support Connectivity Requirements**.
4. Recommend protecting floors while moving heavy pieces of equipment (e.g. ACGD/PDU Cabinet, TRM Body Coil Assembly, TSCC or TGWC).

2–8 FLOOR LOADING AND WEIGHTS

This section contains loading considerations for Signa 1.5T TwinSpeed upgrade components. Listed in Table 2–7 are the weights, floor loading, and normal mounting methods for components.

TABLE 2–7
SIGNA 1.5T TwinSpeed FLOOR LOADING

COMPONENT	NET WT lbs (kg) See Note 2	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
LCC Magnet, Quiet Technology Enclosure, and Coils	See Notes 1 & 3 & Refer to Section 2–8–1	83.27 x 117.67 x 101.96 (2115 x 2989 x 2587.21)	See Illustration 2–7	Magnet & Table Dock Asm. Resting on base, bolted to floor with RF Screen Room vendor supplied/ installed bolts & anchors.
Patient Table (See Note 4)	630 (286)	27 x 89 x 38 (686 x 2261 x 965)	Rectangular base 23 x 65 (584 x 1651). Four casters 8 (203) dia. x 1.5 (38) wide.	Mobile.
Existing Blower Box modified	47 (21)	16 x 16 x 22 (406 x 406 x 559)	Square base 15 x 15 (381 x 381)	Anchor to floor, shelf, or wall.
New Blower Box added	20 (9)	14.5 x 14 x 13.7 (368 x 356 x 348)	Square base 14.5 x 14 (368 x 356)	Anchor to floor or shelf.
SRF Cabinet	525 (239)	24 x 40 x 70.7 (610 x 1016 x 1793)	Rectangular base 24 x 30 (610 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
Signa 1.5 Twin*Speed* with EXCITE Technology System Cabinet	468 (213)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
Signa 1.5 Twin*Speed* System Cabinet	494 (225)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
ACGD/PDU Cabinet	1844 (838)	23.5 x 37.25 x 85.6 (597 x 927 x 2174) See Note 5	Rectangular base 33.5 x 22 (851 x 559). Four casters each 3.0 (76.2) dia. x 1.82 (46.2) wide.	Set on floor on casters.
Twin Accessory Cabinet	600 (273)	23.32 x 38.75 x 50.13 (592 x 984 x 1273)	Rectangular base 22 x 30 (584 x 762). Four casters 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor and rest on casters.
Shield/Cryo Cooler Compressor Cabinet	275 (125)	17.72 x 19.7 x 26.93 (450 x 500 x 684)	Rectangular base 17.72 x 19.7 (450 x 500). Four casters 3 (76) dia. x .75 (19) wide.	Set on floor and rest on casters.
Twin System Cooling Cabinet (TSCC) See Note 6				
Outdoor Air Cooled	1200 (544)	74.83 x 45.5 x 66.88 (1901 x 1156 x 1699)	See Illustration 2–10.	Bolted to mounting pad external to building.
Indoor Water Cooled	1000 (454)	55.4 x 33.59 x 73 (1407 x 358 x 1854)	See Illustration 2–13.	Set on floor and rest on casters.
Indoor Air Cooled	1410 (640)	65 x 32.5 x 71.26 (1651 x 825.5 x 1810)	See Illustration 2–15.	Set on floor and rest on casters.

(Continued)

TABLE 2–7 (Continued)
SIGNA 1.5T TwinSpeed FLOOR LOADING

COMPONENT	NET WT lbs (kg) See Note 2	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
Twin Gradient Water Cooler (TGWC) See Note 6				
Outdoor Air Cooled	780 (354)	49.5 x 34 x 76.2 (1257 x 864 x 1936)	See Illustration 2–17.	Bolted to mounting pad external to building.
Indoor Water Cooled	175 (79)	33.94 x 19.66 x 23.67 (86.21 x 49.94 x 60.13)	See Illustration 2–20.	Set on floor and rest on 4 feet.
Indoor Air Cooled	780 (354)	28.4 x 33 x 73 (721 x 838 x 1854)	See Illustration 2–22.	Set on floor and rest on casters & front pads on anti–tip legs
*Advantage Workstation	Refer to Direction *2261309–100 Advantage Workstation Ultra60 Pre–Installation.* Also refer to **Section 2–6–3 Advantage Workstation Additional Hardware Features** to determine location of workstation.			
MNS Cabinet Option	430 (195)	23.56 x 42 x 70.7 (598 x 1067 x 1797)	Rectangular base 33.5 x 22 (851 x 559).	Casters for location. Set on floor on four leveling pads.

Note 1 Weight of 1.5T LCC (Active Shield) magnet with Quiet Technology Enclosure, RF/Gradient (TRM) Body Coil, and cryogens is 12,826 lbs (5818 kg).

2 Consult a structural engineer on method of calculating proper weight/unit area for floor loading.

3 Refer to **Section 2–3 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY** for Gradient Coil Assembly replacement weight and dimension requirements.

4 630 lbs (286 kg) includes 350 lbs (159 kg) patient.

5 ACGD/PDU Cabinet height with Fan Module removed is 74.8 inches (1900 mm).

6 System Cooling Cabinet is available in the following configuration options (must choose one):
TSCC (Provides water cooling for Shield/Cryo Cooler Compressor and Gradient Coil).
- Outdoor Air Cooled TSCC installed external to building with Remote Control Panel (RCP) wall mounted in the Signa system Equipment Room
- Indoor Water Cooled TSCC located in or near the Signa system Equipment Room
- Indoor Air Cooled TSCC located in or near the Signa system Equipment Room.
- True Mobile system equipment, contact Van Manufacturer for details.

TGWC (Provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor, refer to Section 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.
- Outdoor Air Cooled TGWC installed external to building with Remote Control Panel (RCP) wall mounted in the Signa system Equipment Room.
- Indoor Water Cooled TGWC and Remote Control Panel (RCP) wall mounted both located in the Signa system Equipment Room
- Indoor Air Cooled TGWC located in or near the Signa system Equipment Room.

7 The Operator Workspace Cabinet minimum area location is under the Workspace Table, see in Illustration 2–40. An alternate location is possible with the Operator Workspace Cabinet located against the right side of the Workspace Table.

* Optional Equipment.

2–8–1 Magnet Loading Considerations

Refer to ***Direction 2294003 Signa 1.5T TwinSpeed Pre–Installation*** **Section 2–10–1 Magnet Loading Considerations** for Signa 1.5T TwinSpeed LCC Magnet information.

2–8–2 Operator Workspace Mounting Requirements

The Operator Workspace Table sets on the floor with the Workspace Cabinet positioned under the Table towards the right side, see Illustration 2–40. Note, the Workspace Table must be bolted to the floor for safe use of the table and equipment.

2–9 COMPONENT DIMENSIONS

To assist in completing your room layout, refer to Table 2–8 for list of component Illustrations and the upgrade catalog numbers which provide the new equipment. Included are items which may impact site layout or need to be taken into consideration when determining delivery route. Modifications to existing cabinets are not included, catalogs may list the same new equipment but differ due to other modification hardware.

TABLE 2–8
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M1000MN M1000MP M1000MR M1000MS M1000MW M1000NH M1000NJ M1000NK M1000NT	OPERATOR WORKSPACE (OW1)	2–40
M1033K M1033L	OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – OCTANE COMPUTER	2–43
M1033MD	MNS CABINET (MNS) – 1.5T OPTION	2–45
M1060JW	LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)	2–9
M1090TL	PATIENT TRANSPORT TABLE (PT1)	2–32
M3000TA	PATIENT TRANSPORT TABLE (PT1) TWIN ACCESSORY CABINET (TAC) ACGD/PDU CABINET (MR3) SRF CABINET (MR1) SYSTEM CABINET (MR2) FOR SIGNA TWIN*SPEED* OPERATOR WORKSPACE (OW1)	2–32 2–35 2–36 2–37 2–39 2–40
M3000TC	PATIENT TRANSPORT TABLE (PT1) TWIN ACCESSORY CABINET (TAC) ACGD/PDU CABINET (MR3) SRF CABINET (MR1) SYSTEM CABINET (MR2) FOR SIGNA TWIN*SPEED* WITH EXCITE TECHNOLOGY OPERATOR WORKSPACE (OW1)	2–32 2–35 2–36 2–37 2–38 2–40
M3043TA M3044TA M3063TA M3043TC M3044TC M3063TC	UPGRADE NEW BLOWER BOX (MG6) PENETRATION PANEL (PP1) PENETRATION PANEL COVER SPT PHANTOM SET SHIPPING/STORAGE CABINET	2–33 2–46 2–47 2–48
M3060TA	SIGNA TWIN*SPEED* 1.5T LCC (ACTIVE SHIELD) MAGNET ENCLOSURE (FRONT AND REAR VIEWS) SIGNA TWIN*SPEED* 1.5T LCC (ACTIVE SHIELD) MAGNET CABLE ACCESS 1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN MAGNET MONITOR (MSM1)	2–4 2–5 2–7 2–8
	(Continued)	

2–9 COMPONENT DIMENSIONS (Continued)

TABLE 2–8 (Continued)
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M3085TA	UPGRADE TRM GRADIENT COIL & CRADLE ON CART **Note:** Systems installing M1060TA 1.5T Magnet With TwinSpeed Enclosure will receive the magnet with the TRM installed in the magnet bore.	2–2
M3088TA	OUTDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT/O] (TSCC)	2–10
	OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O] MOUNTING	2–11
	REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O]	2–12
M3088TB	INDOOR WATER COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 18MT/WC] (TSCC) FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER	2–13
	INDOOR WATER COOLED TSCC [MODEL LC 18MT/WC] MOUNTING	2–14
M3088TC	INDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT] (TSCC) FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER	2–15
	INDOOR AIR COOLED TSCC MOUNTING	2–16
M3088TE	INDOOR WATER COOLED TWIN GRADIENT WATER COOLER [MODEL LTL 10] (TGWC)	2–20
	REMOTE CONTROL PANEL (RCP) FOR INDOOR WATER COOLED TGWC [MODEL LTL 10]	2–21
M3088TF	OUTDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10M/O] (TGWC)	2–17
	OUTDOOR AIR COOLED TGWC [MODEL LC 10M/O] MOUNTING	2–18
	REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TGWC [MODEL LC–10M/O]	2–19
M3088TG	INDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10M] (TGWC)	2–22
M3088TH	SIGNA TWIN*SPEED* MAIN DISCONNECT PANEL (MDP) M3088TH	2–49
M3090AH	ACGD/PDU CABINET (MR3)	2–36
	SRF CABINET (MR1)	2–37
	SYSTEM CABINET (MR2)	2–39
	OPERATOR WORKSPACE (OW1)	2–40
M3090BB	ACGD/PDU CABINET (MR3)	2–36
	SRF CABINET (MR1)	2–37
M3090BC	ACGD/PDU CABINET (MR3)	2–36
	SRF CABINET (MR1)	2–37
M3090DB M3090DC	MNS CABINET (MNS) – 1.5T OPTION	2–45
M3090EC	SYSTEM CABINET (MR2) FOR SIGNA TWIN*SPEED* WITH EXCITE TECHNOLOGY	2–38
M3090ED	SRF CABINET (MR1)	2–37
	SYSTEM CABINET (MR2) FOR SIGNA TWIN*SPEED* WITH EXCITE TECHNOLOGY	2–38
	OPERATOR WORKSPACE (OW1)	2–40
M3090TA	TWIN ACCESSORY CABINET (TAC)	2–35
	REAR PEDESTAL	*See Note 1*
M3090TE	UPGRADE MODIFIED BLOWER BOX (MG6) *See Note 2*	2–34
	TWIN ACCESSORY CABINET (TAC)	2–35
(Continued)		

2–9 COMPONENT DIMENSIONS (Continued)

TABLE 2–8 (Continued)
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

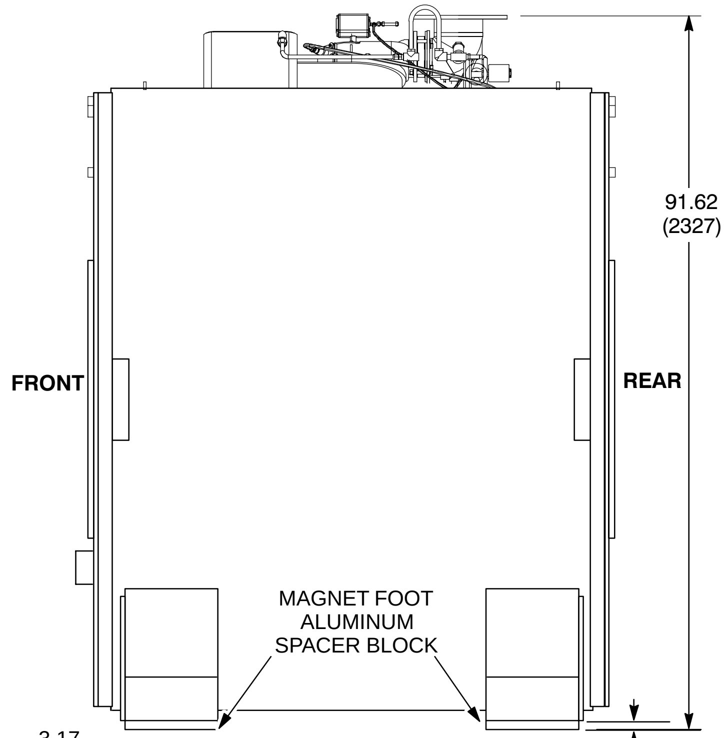
UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M3090TF	FRONT RIGHT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2–23
	FRONT LEFT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2–24
	REAR RIGHT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2–25
	REAR LEFT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2–26
	FRONT ENDBELL ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2–27
	FRONT COSMETIC COVER ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2–28
	REAR ENDBELL ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2–29
	REAR COSMETIC COVER ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE	2–30
M3090TJ	FRONT SPLIT BRIDGE	2–31
M3090TM	TWIN ACCESSORY CABINET (TAC)	2–35

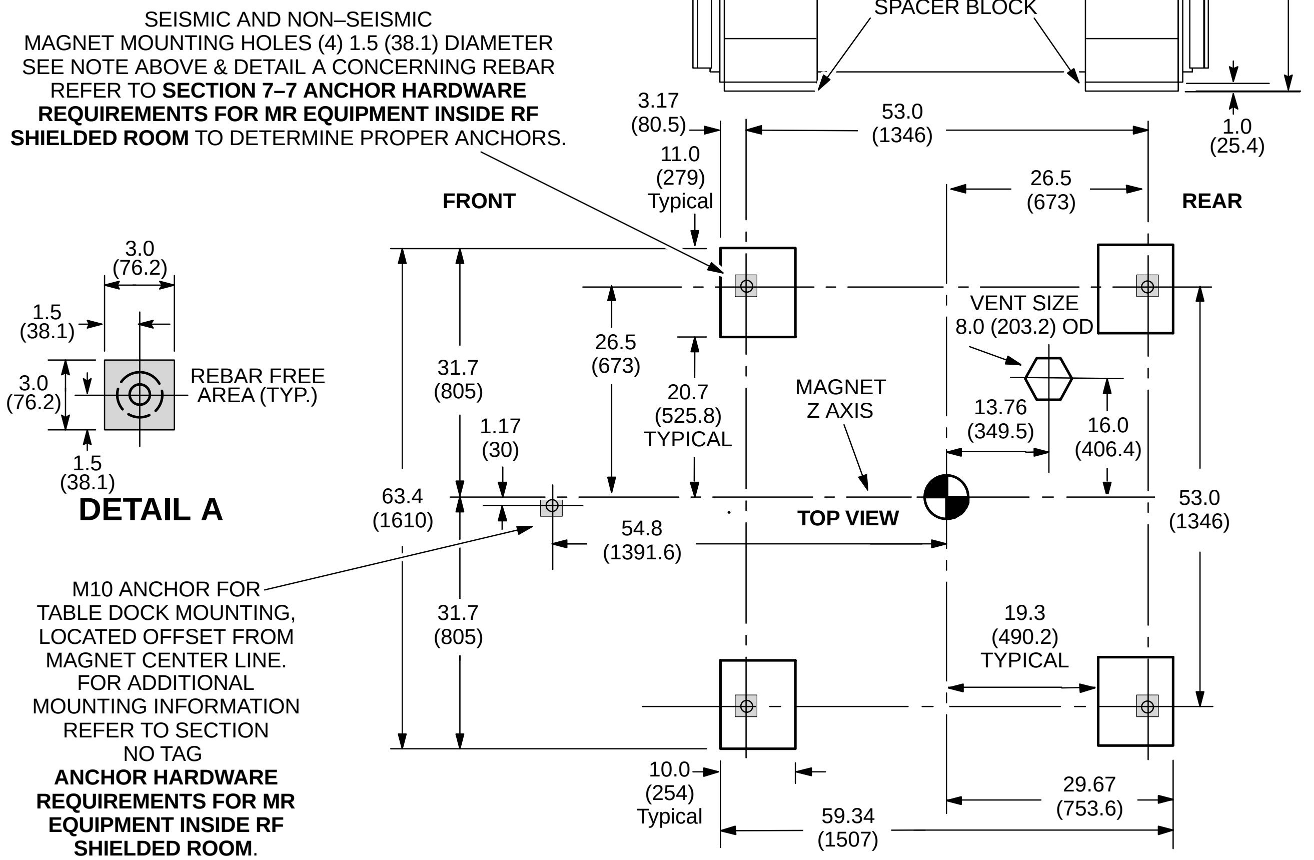
Note 1 Rear Pedestal is supplied with the indicated catalog numbers. Refer to Section 2–3, MINIMUM DOOR/HALLWAY SIZES, for dimension information.

2 Catalog provides a kit to modify existing system equipment to the configuration shown in the illustration.

NOTE:

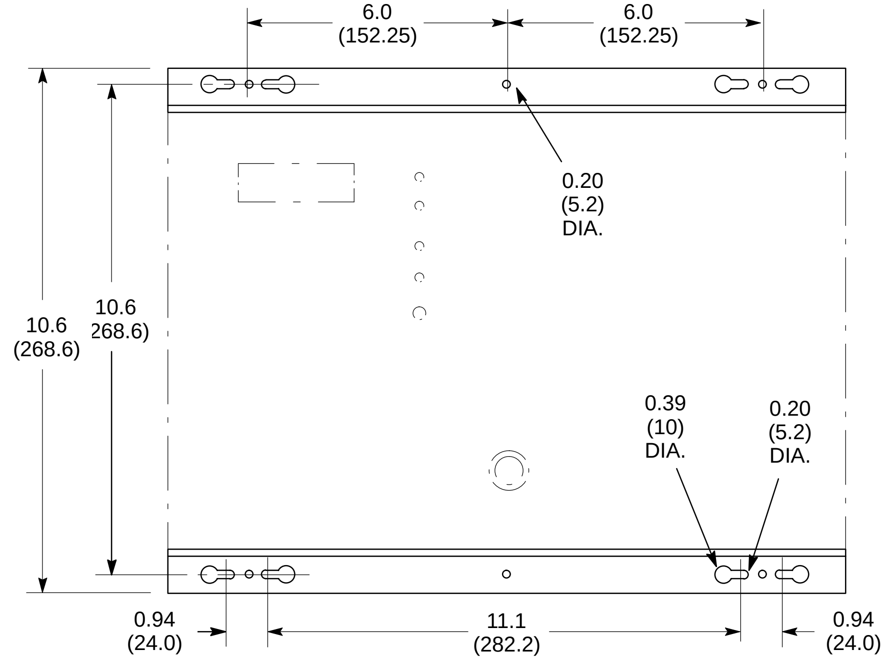
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- LIFTING/JACKING BEAMS ARE ATTACHED TO SIDES OF MAGNET. REFER TO ILLUSTRATION 8–1.
- ALUMINUM LEVELING PLATES ARE NOT SHOWN.
- APPROX. WEIGHT OF MAGNET, CRYOGENS, COILS, AND ENCLOSURE WITH TRM 12,826 lbs (5818 kg)
- STEEL REBAR MUST NOT BE POSITIONED IN SHADED AREAS TO PREVENT INTERFERENCE WITH MOUNTING BOLTS.
- FOR MAGNET MOVING DIMENSIONS REFER TO **SECTION 8 SHIPPING AND DELIVERY DATA**



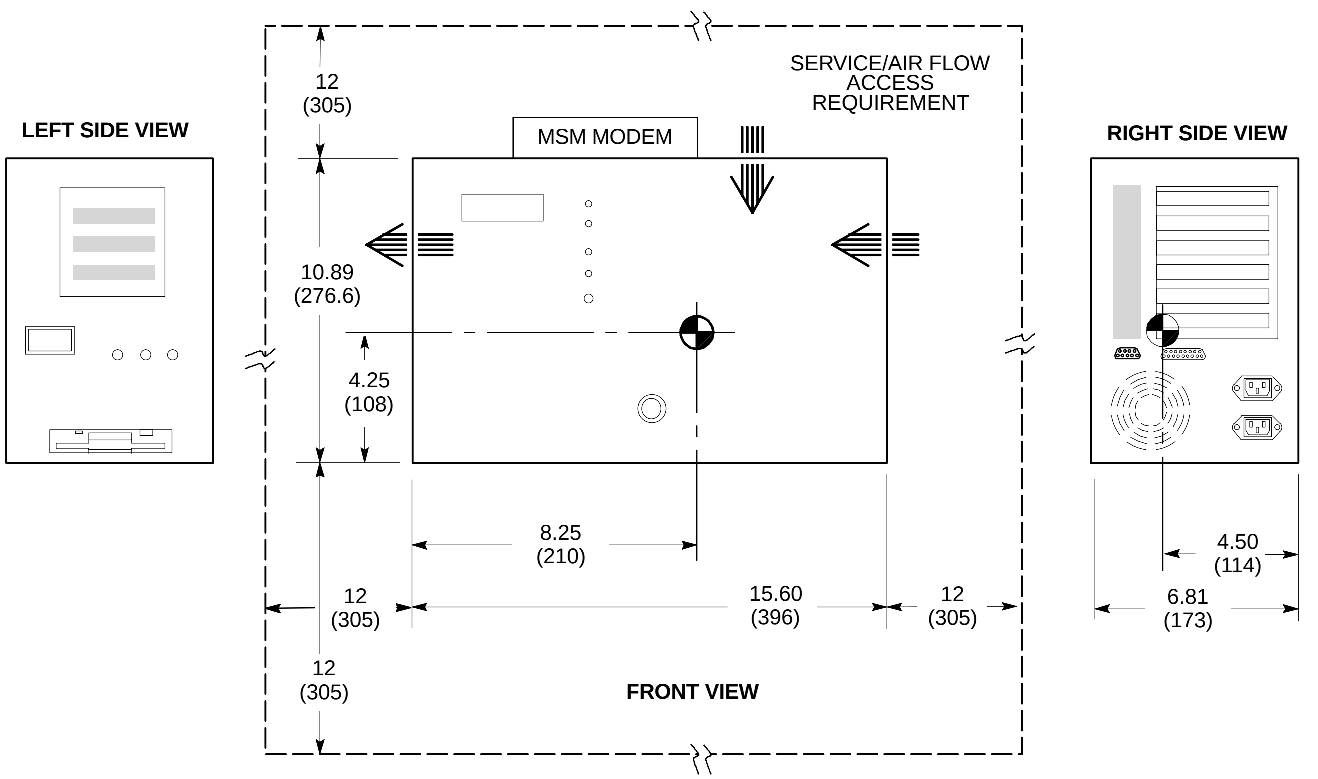


1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN
ILLUSTRATION 2–7

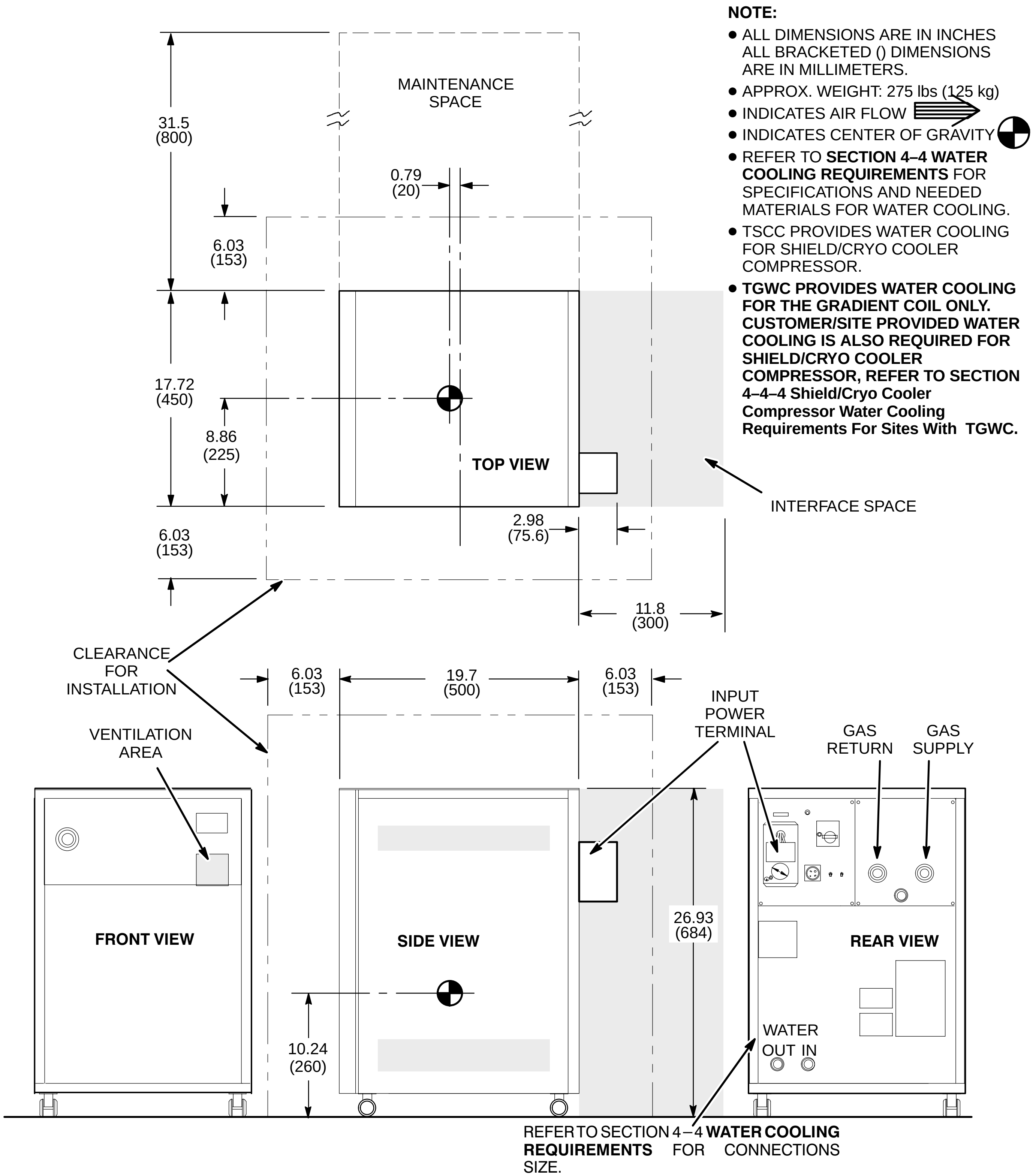
NOTE:
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 22 lbs (10 kg)
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY



MOUNTING PATTERN

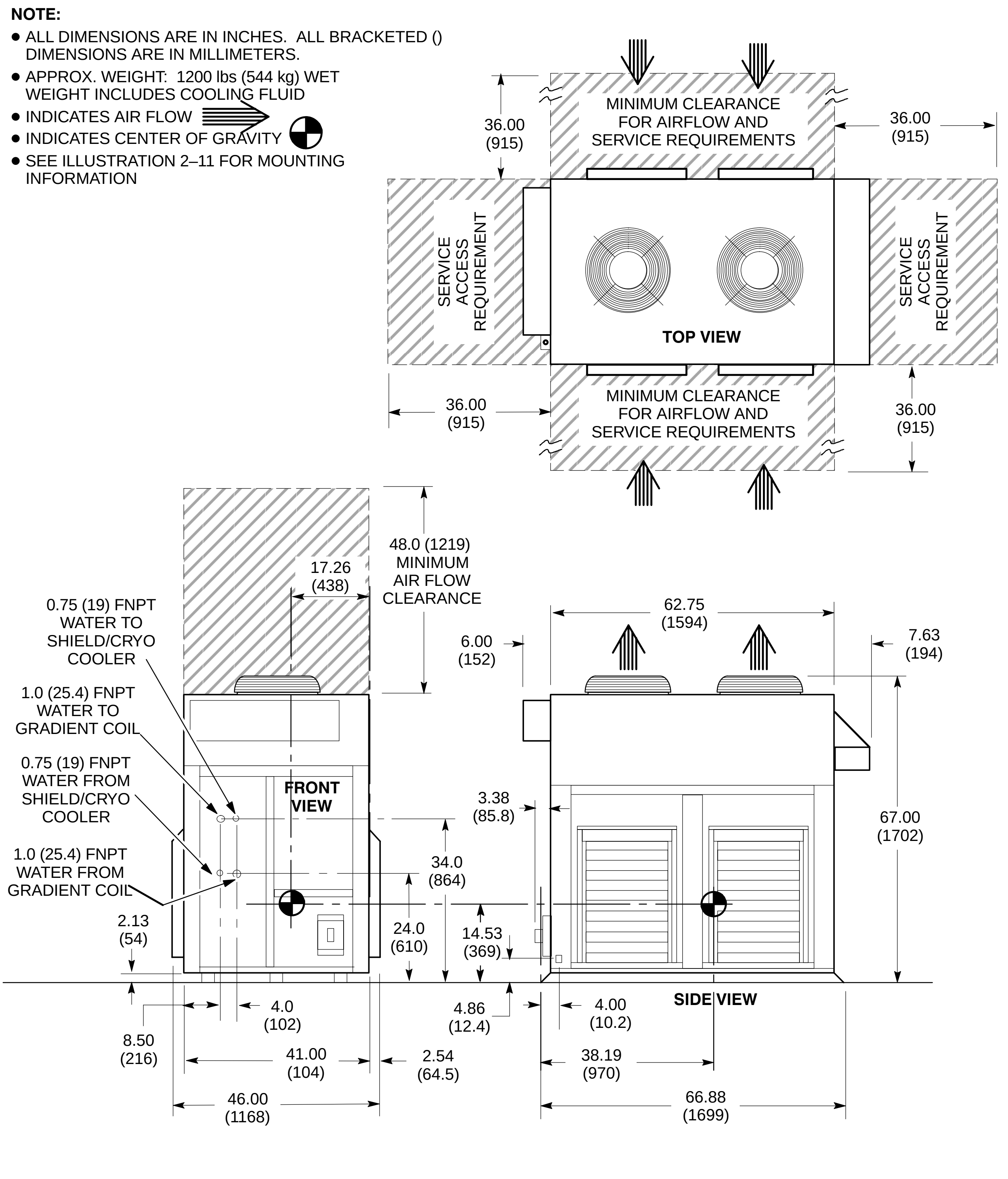


MAGNET MONITOR (MSM1)
ILLUSTRATION 2–8



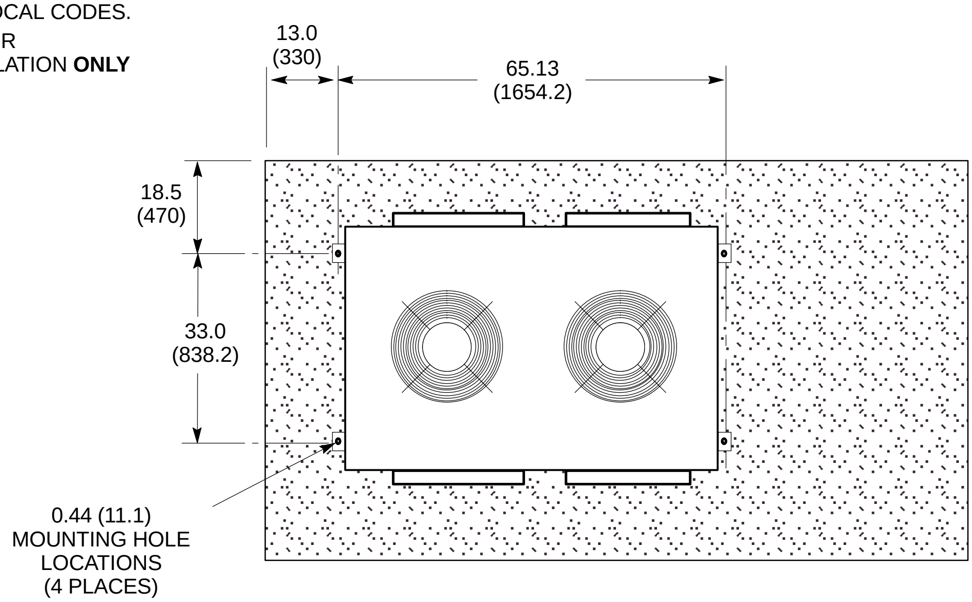
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)

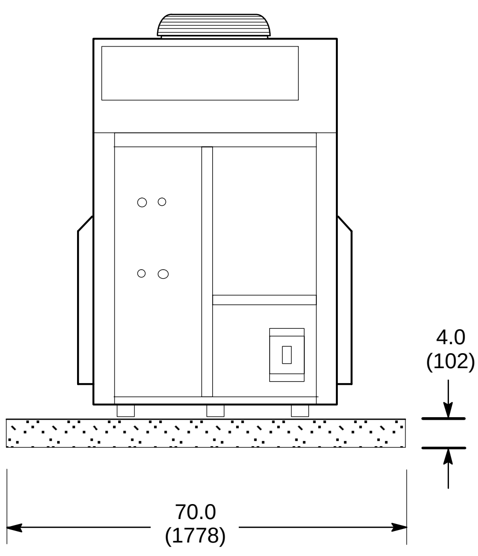
ILLUSTRATION 2–9

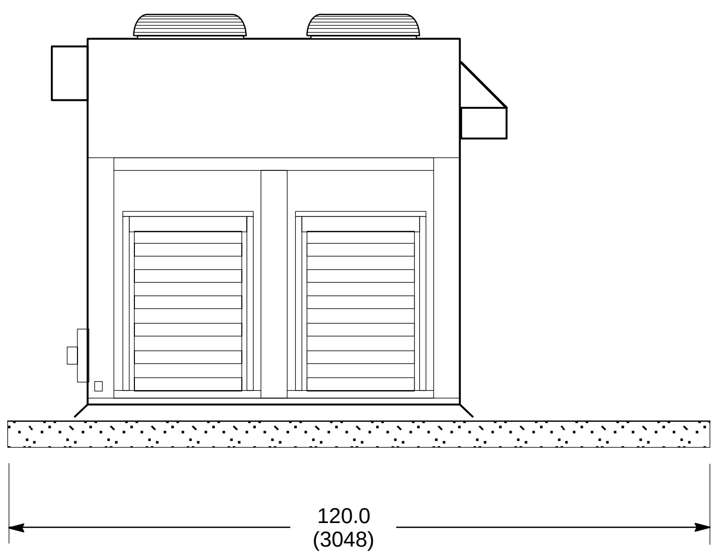


OUTDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT/O] (TSCC)
ILLUSTRATION 2–10

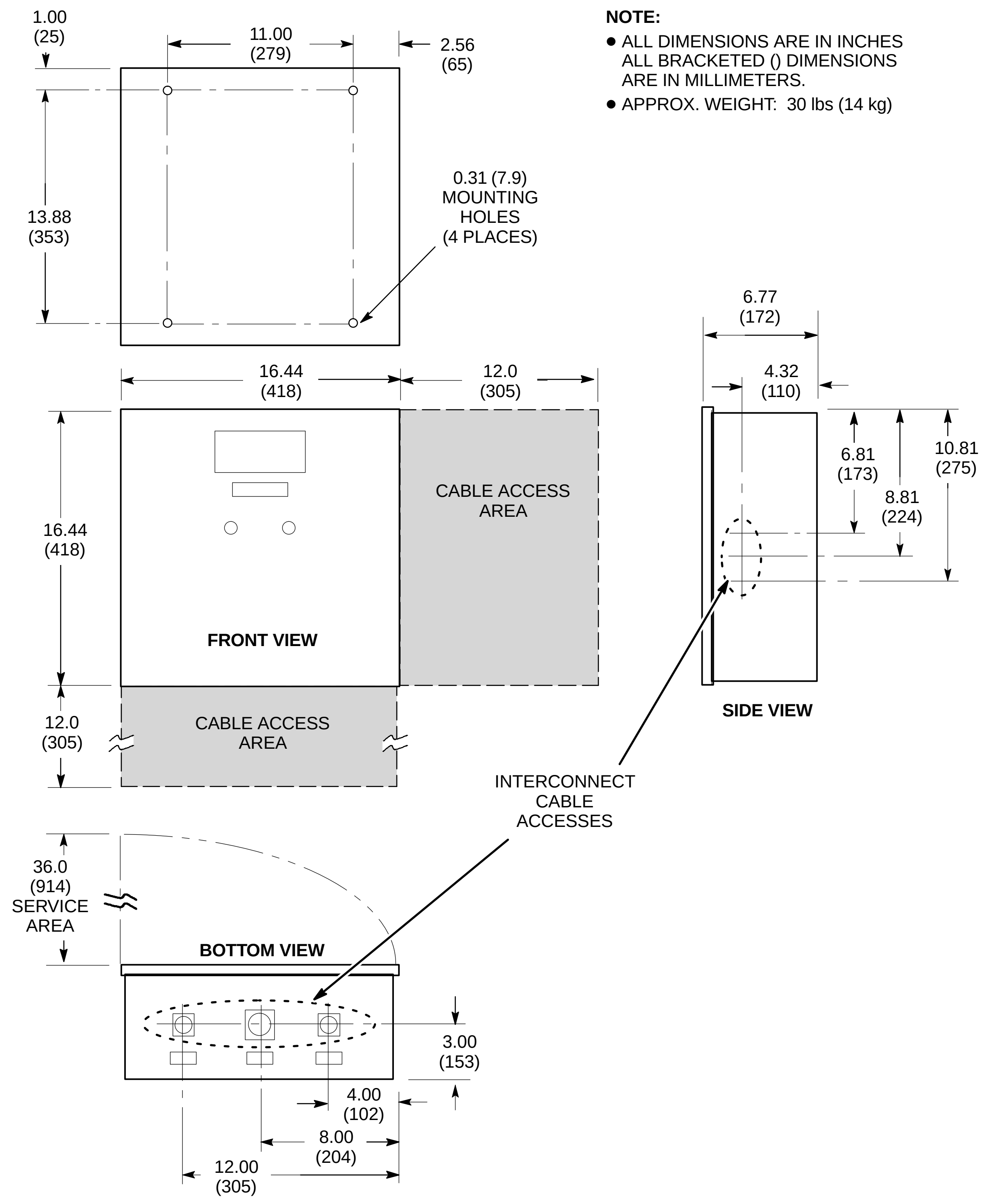
NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MINIMUM 4 INCH (102 MM) DEEP CONCRETE PAD OF 2500 PSI CONCRETE REQUIRED. ADDITIONAL CONCRETE DEPTH MAY BE REQUIRED AS DICTATED BY LOCAL CODES.
- CONCRETE PAD REQUIRED FOR ON–GRADE (GROUND) INSTALLATION **ONLY**



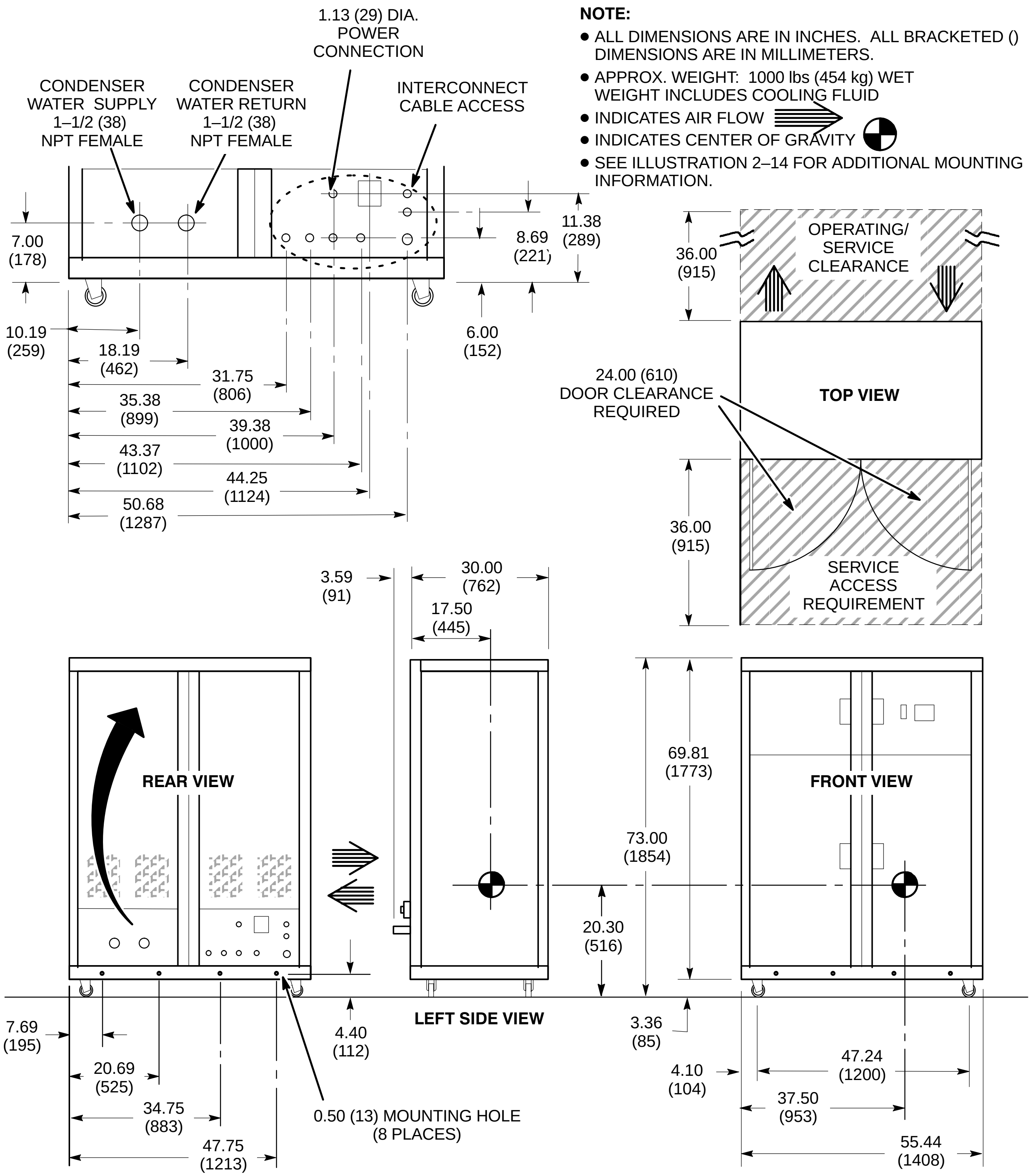




OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O] MOUNTING
ILLUSTRATION 2–11

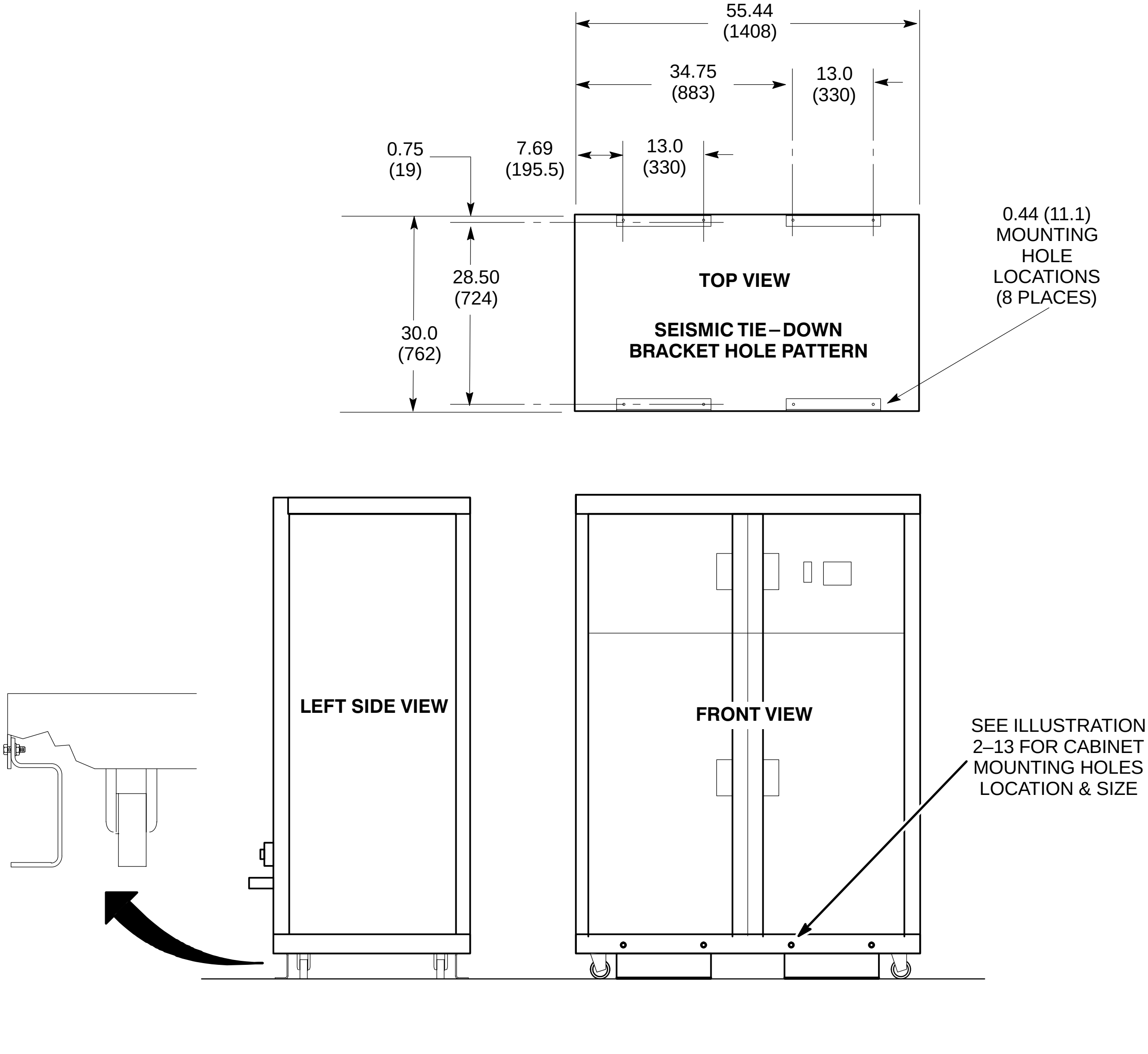


REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O]
ILLUSTRATION 2–12



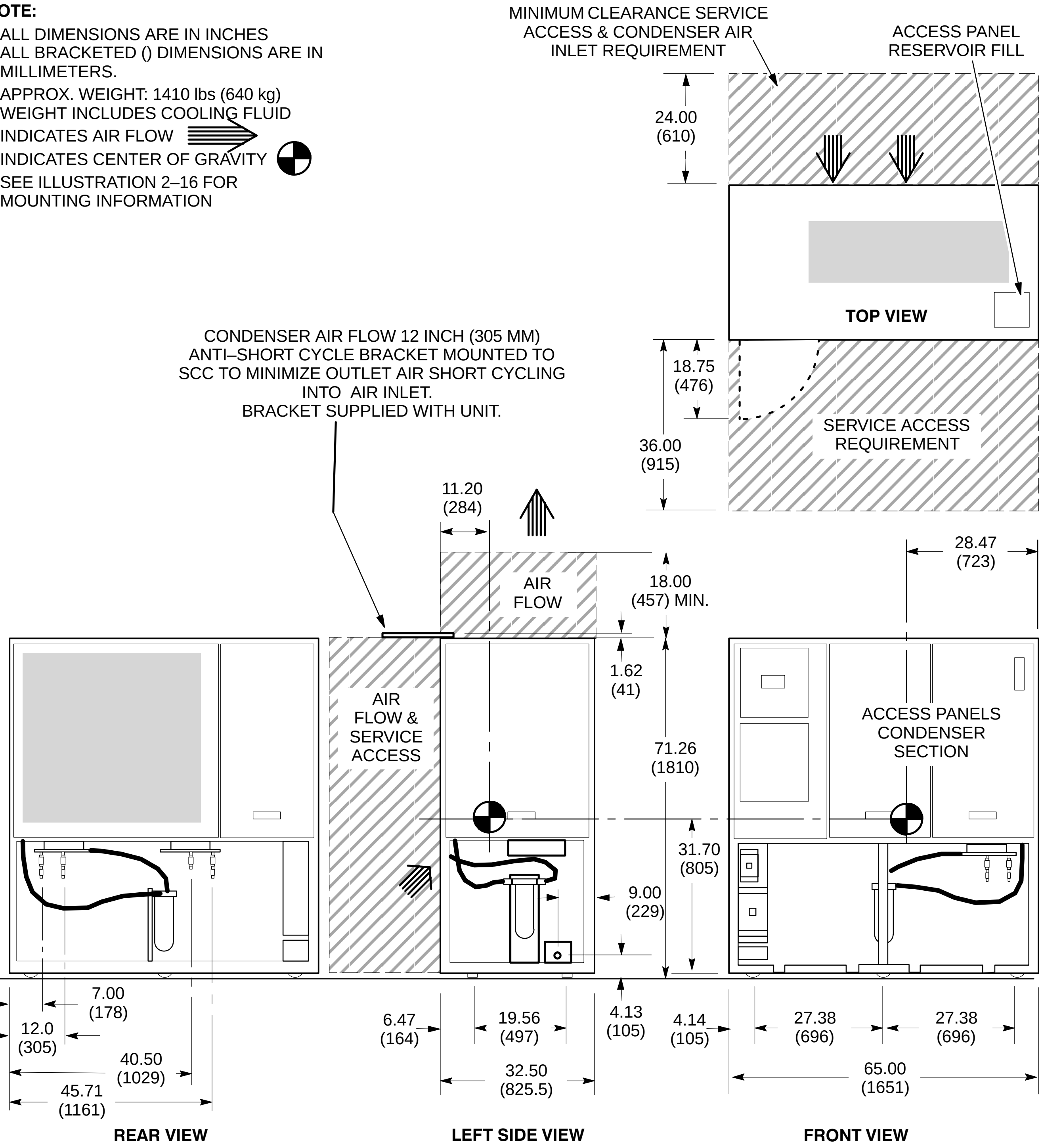
INDOOR WATER COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 18MT/WC] (TSCC)
FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER
ILLUSTRATION 2–13

NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



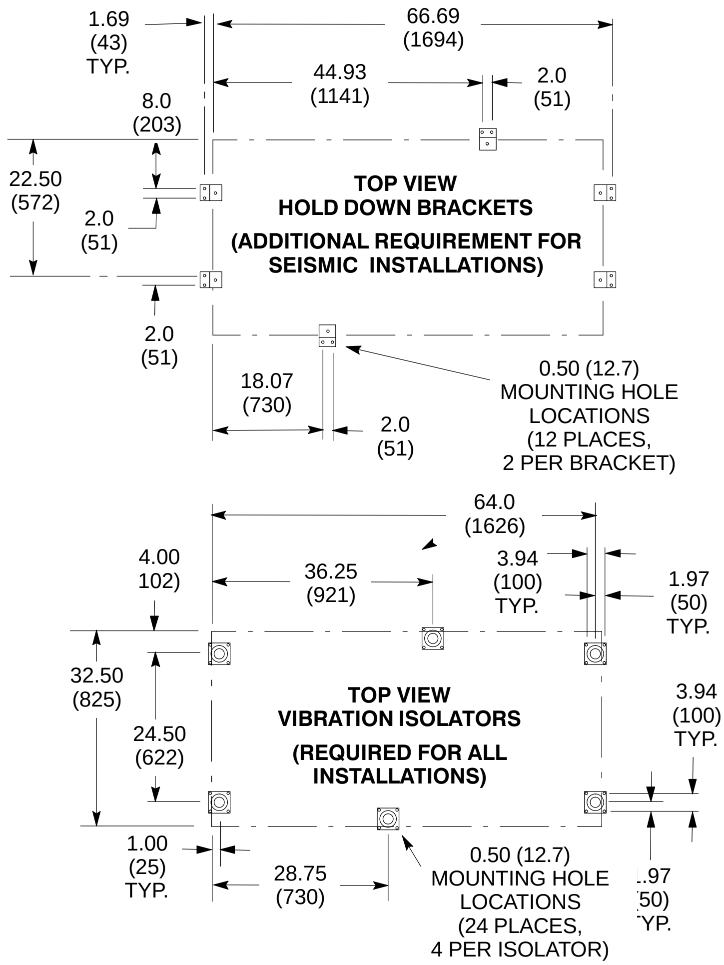
INDOOR WATER COOLED TSCC [MODEL LC 18MT/WC] MOUNTING
ILLUSTRATION 2–14

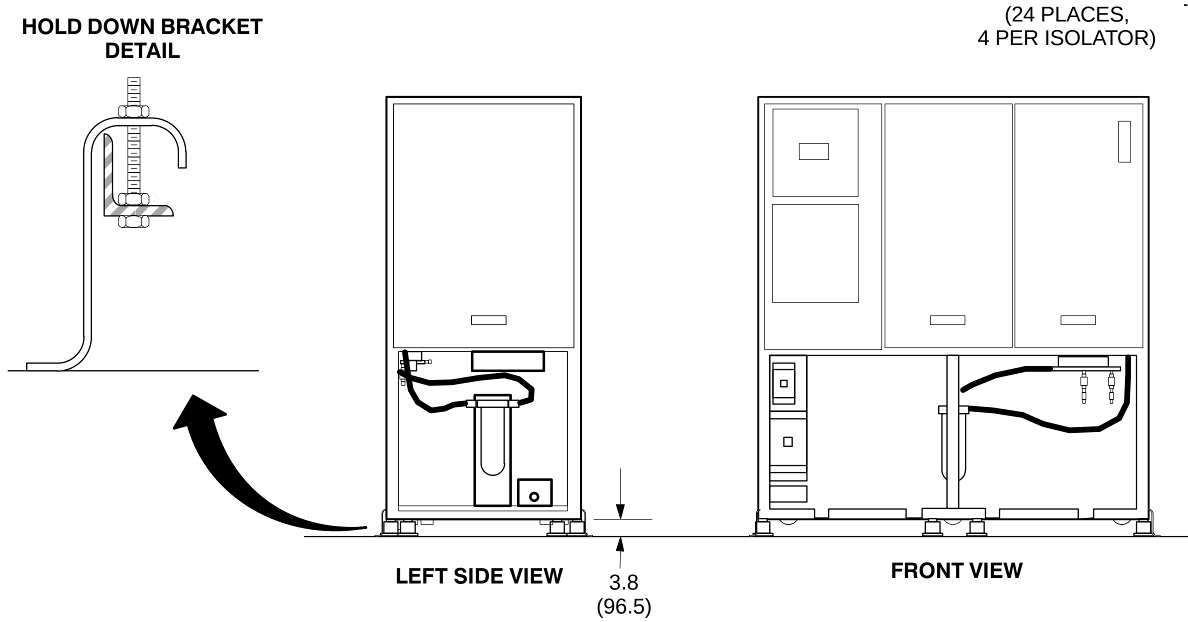
NOTE:
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1410 lbs (640 kg) WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- SEE ILLUSTRATION 2–16 FOR MOUNTING INFORMATION



INDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT] (TSCC)
FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER
ILLUSTRATION 2–15

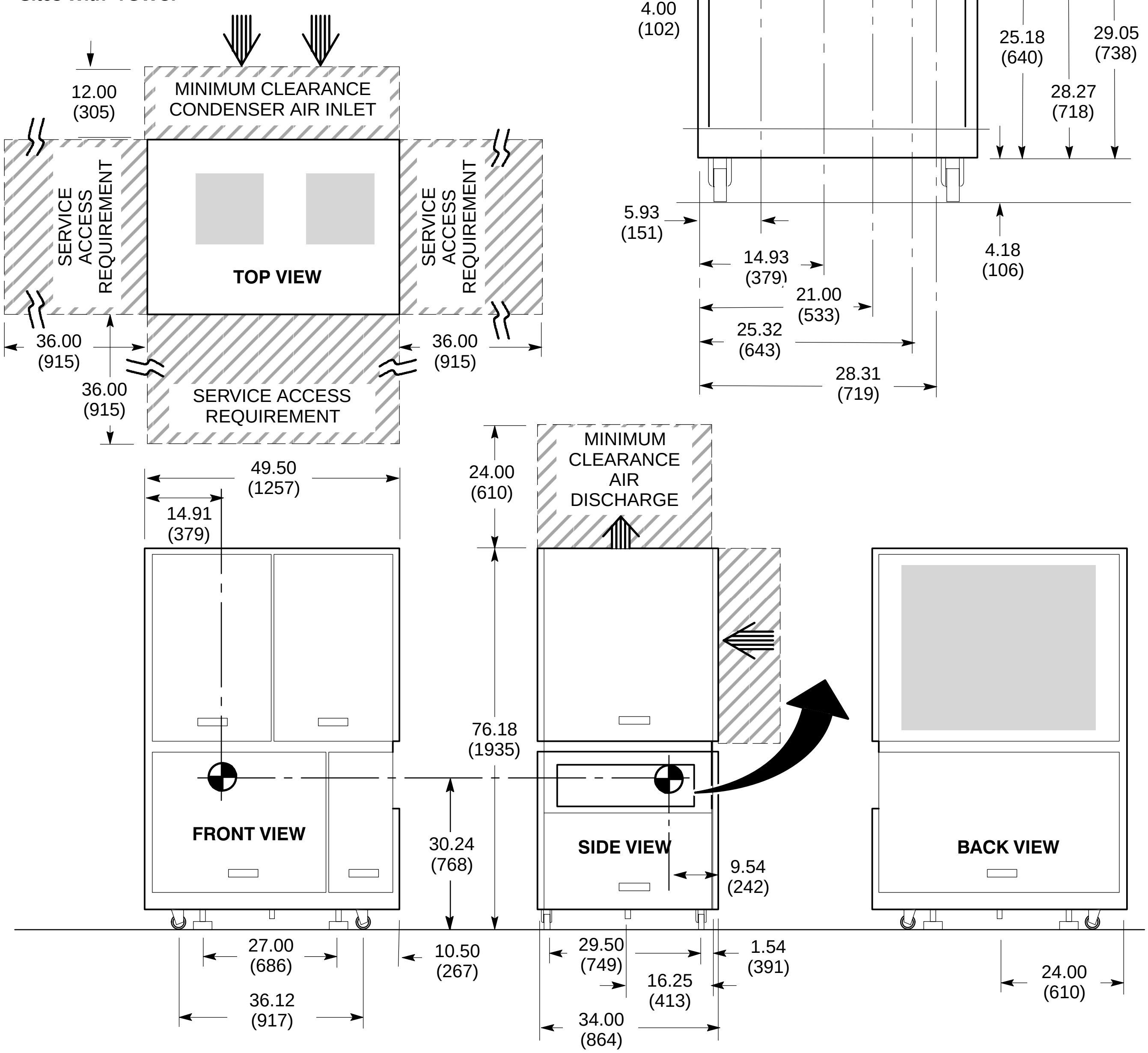
NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.





INDOOR AIR COOLED TSCC MOUNTING
ILLUSTRATION 2–16

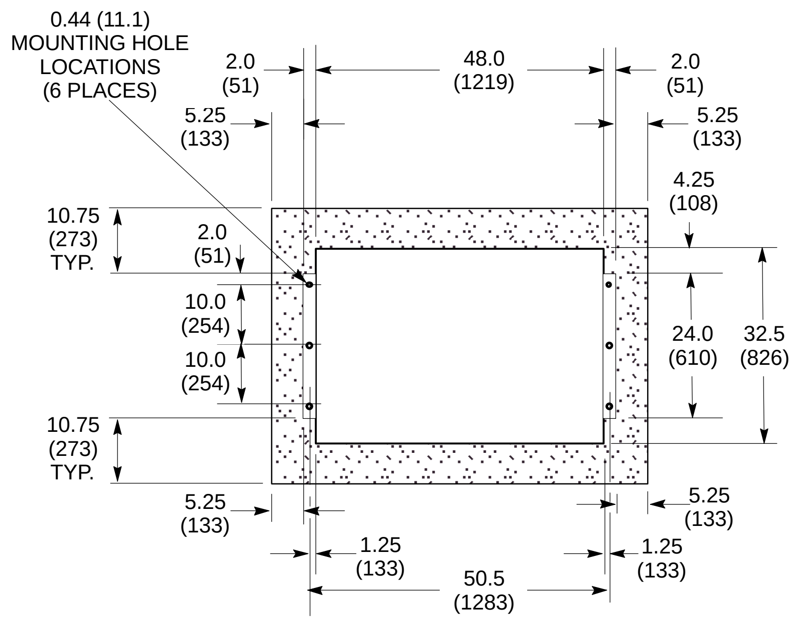
NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 780 lbs (345 kg) WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- SEE ILLUSTRATION 2–18 FOR MOUNTING INFORMATION
- **TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.**



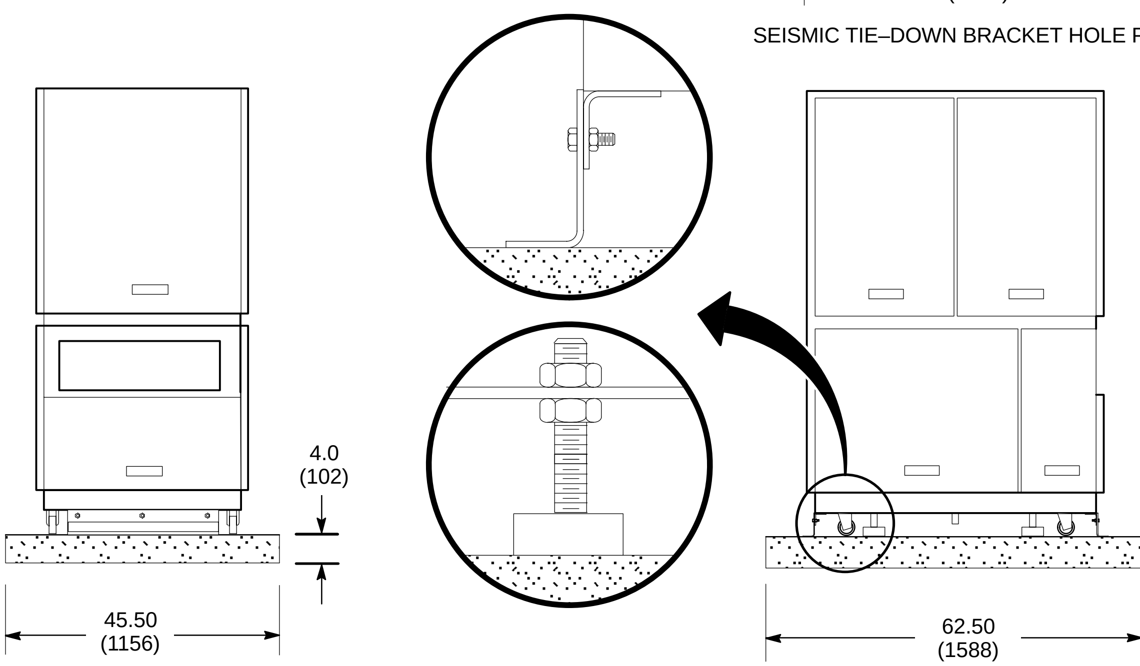
OUTDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10MO] (TGWC)
FOR GRADIENT COIL COOLING WATER ONLY
ILLUSTRATION 2–17

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MINIMUM 4 INCH (102 MM) DEEP CONCRETE PAD OF 2500 PSI CONCRETE REQUIRED. ADDITIONAL CONCRETE DEPTH MAY BE REQUIRED AS DICTATED BY LOCAL CODES.
- CONCRETE PAD REQUIRED FOR ON–GRADE (GROUND) INSTALLATION **ONLY**



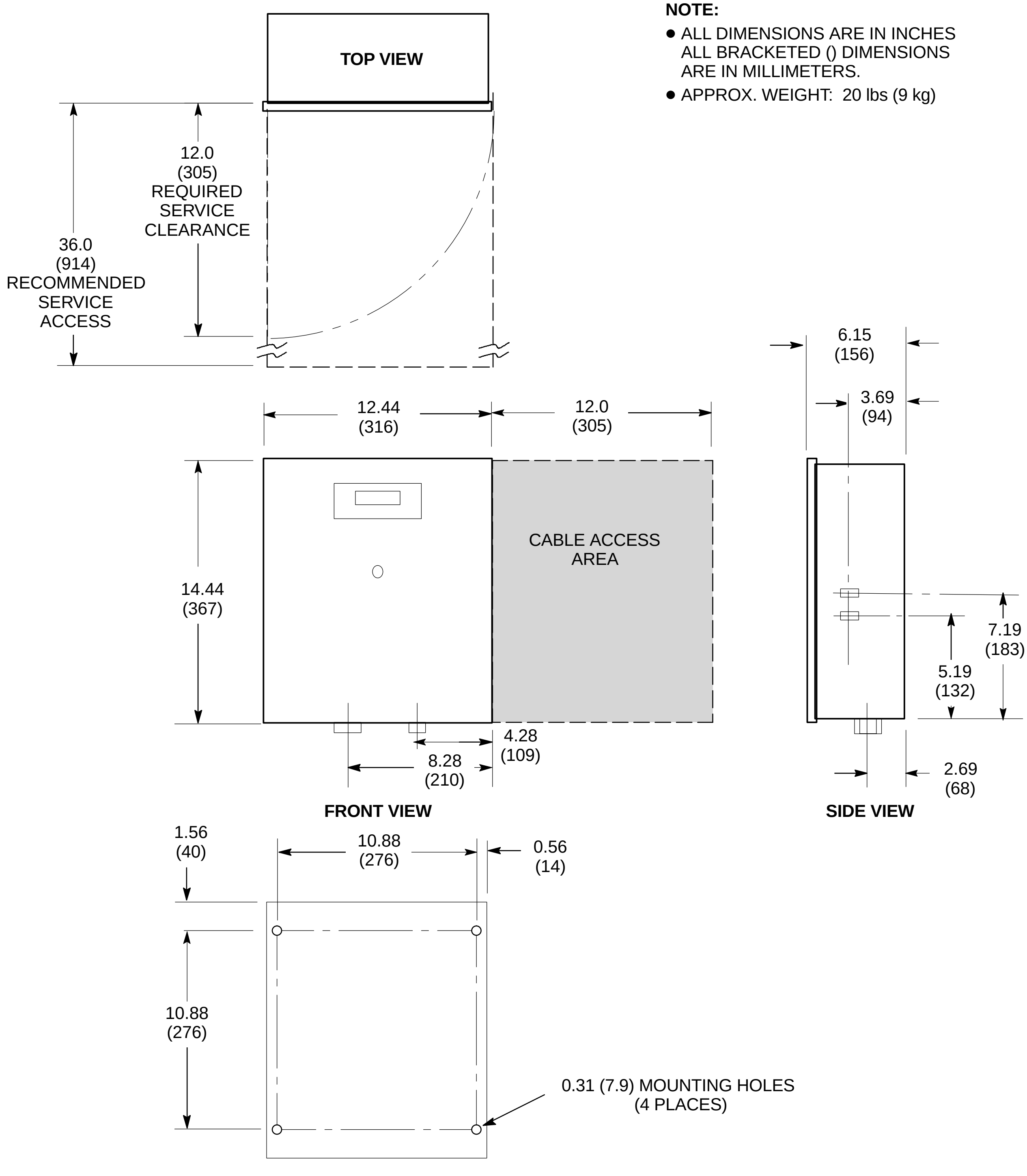
SEISMIC TIE–DOWN BRACKET HOLE PATTERN



RIGHT SIDE VIEW **FRONT VIEW**

OUTDOOR AIR COOLED TGWC [MODEL LC 10M/O] MOUNTING
ILLUSTRATION 2–18

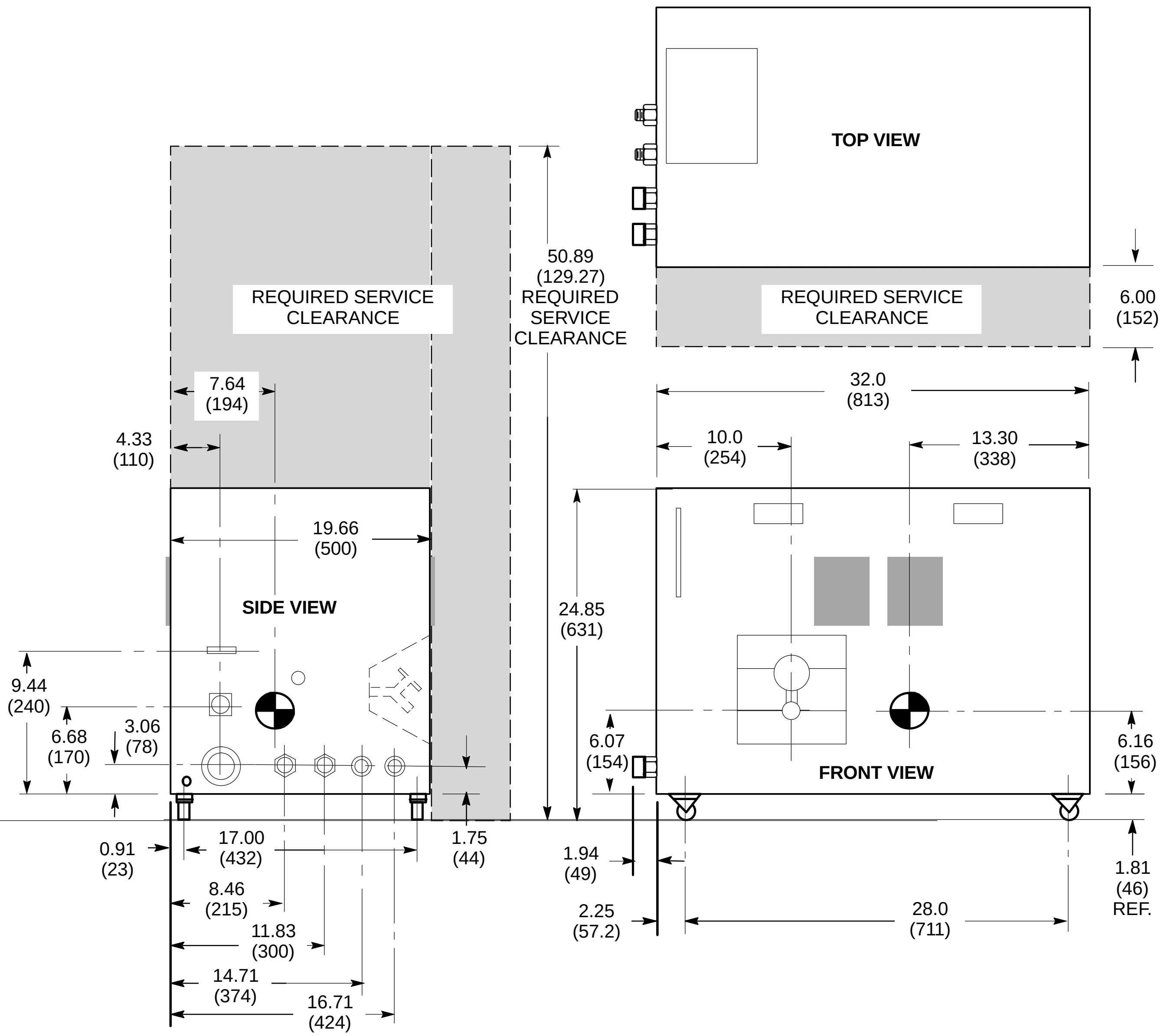
NOTE:
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



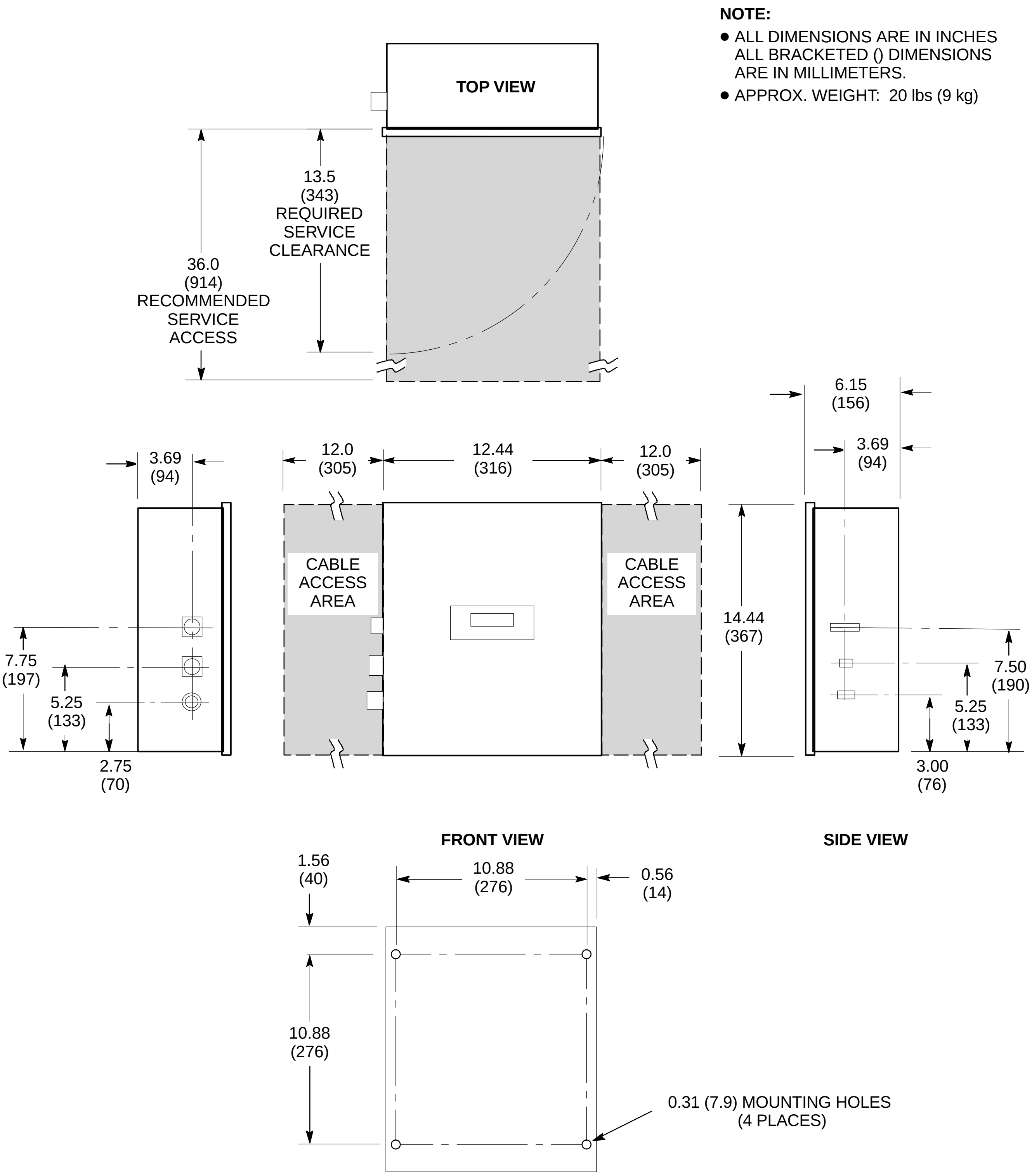
REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TGWC [MODEL LC 10M/O]
ILLUSTRATION 2–19

NOTE:

- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: TGWC 175 lbs (79 kg) WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- **TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.**



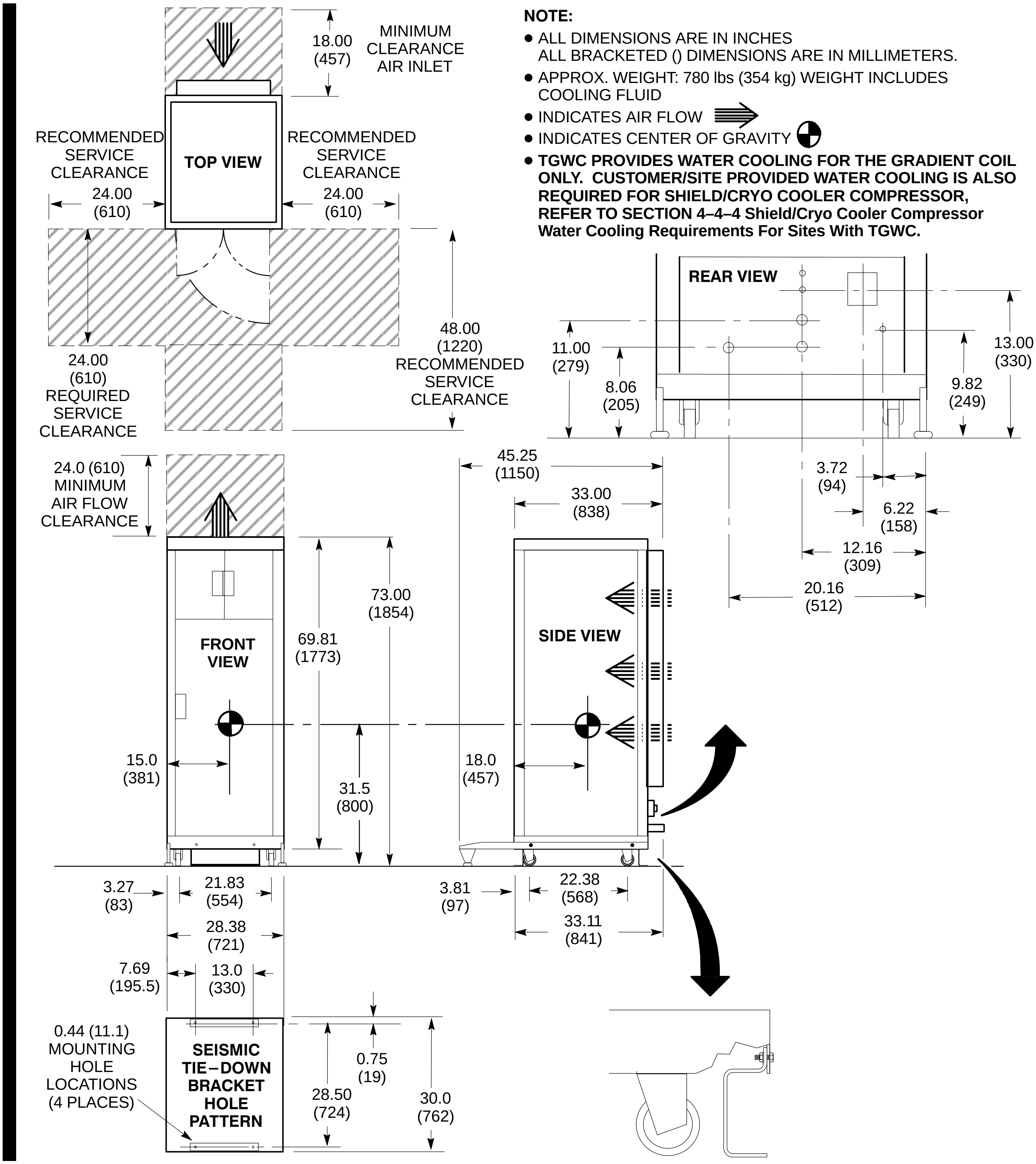
INDOOR WATER COOLED TWIN GRADIENT WATER COOLER [MODEL LTL–10] (TGWC)
ILLUSTRATION 2–20



REMOTE CONTROL PANEL (RCP) FOR INDOOR WATER COOLED TGWC [MODEL LTL–10]
ILLUSTRATION 2–21

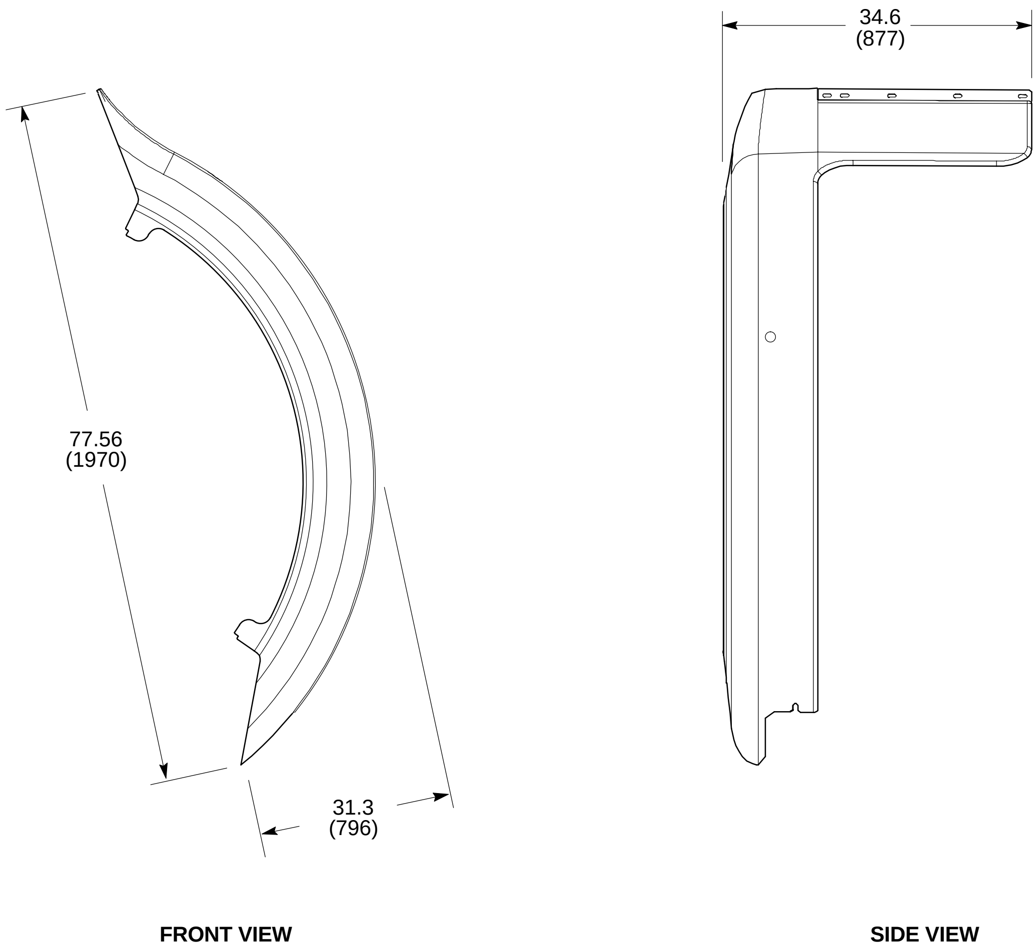
NOTE:

- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 780 lbs (354 kg) WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- **TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.**



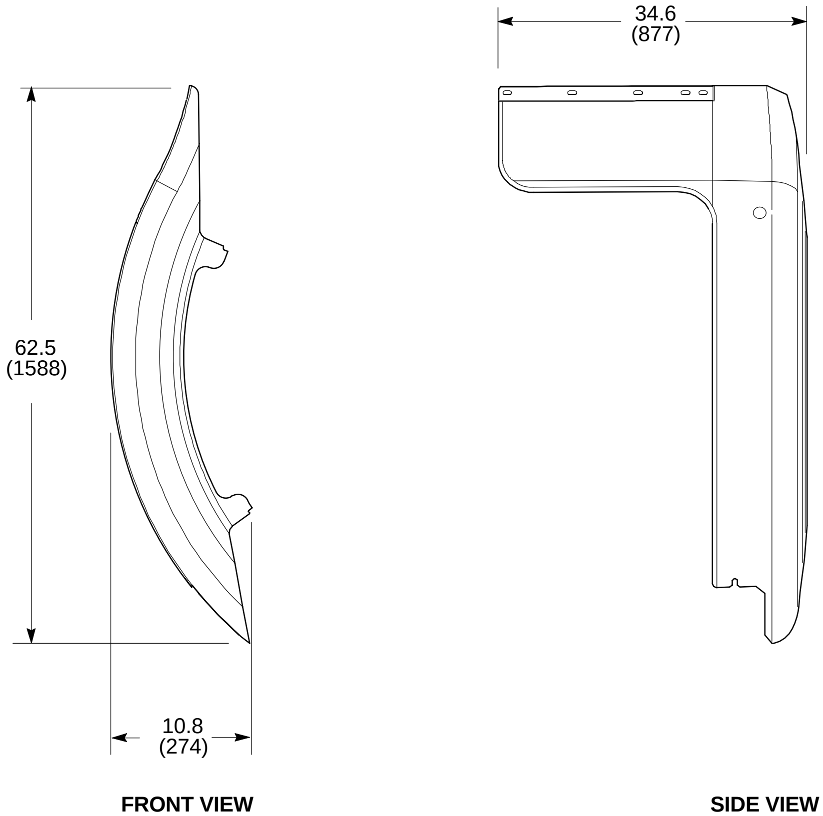
INDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10M] (TGWC)
ILLUSTRATION 2–22

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



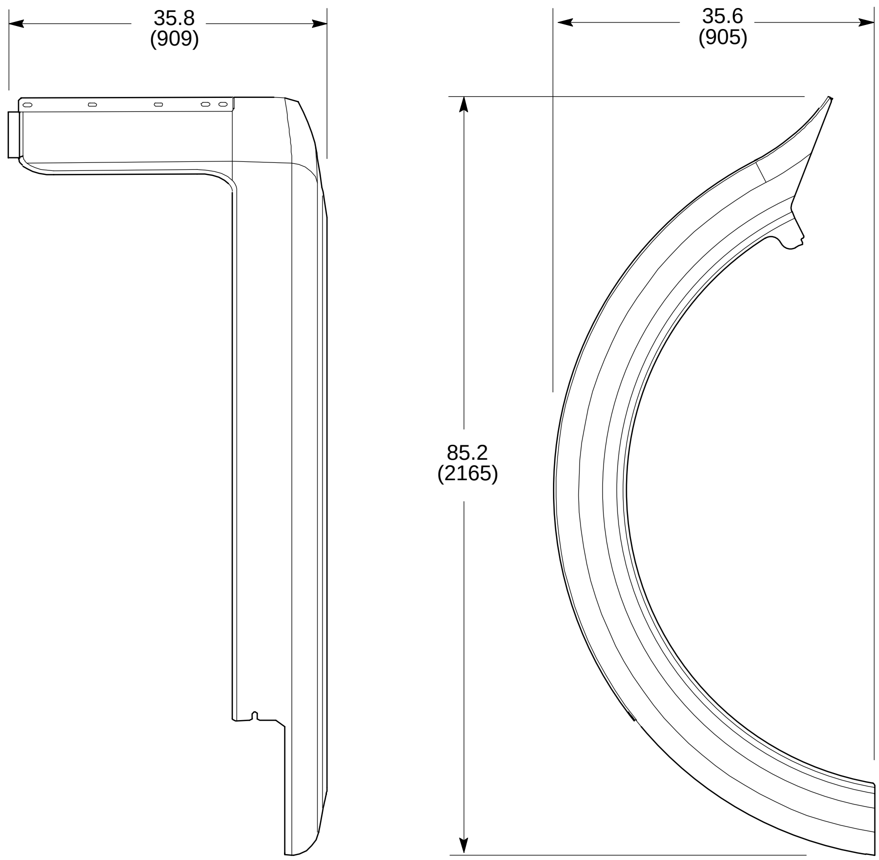
FRONT RIGHT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2–23

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



FRONT LEFT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2–24

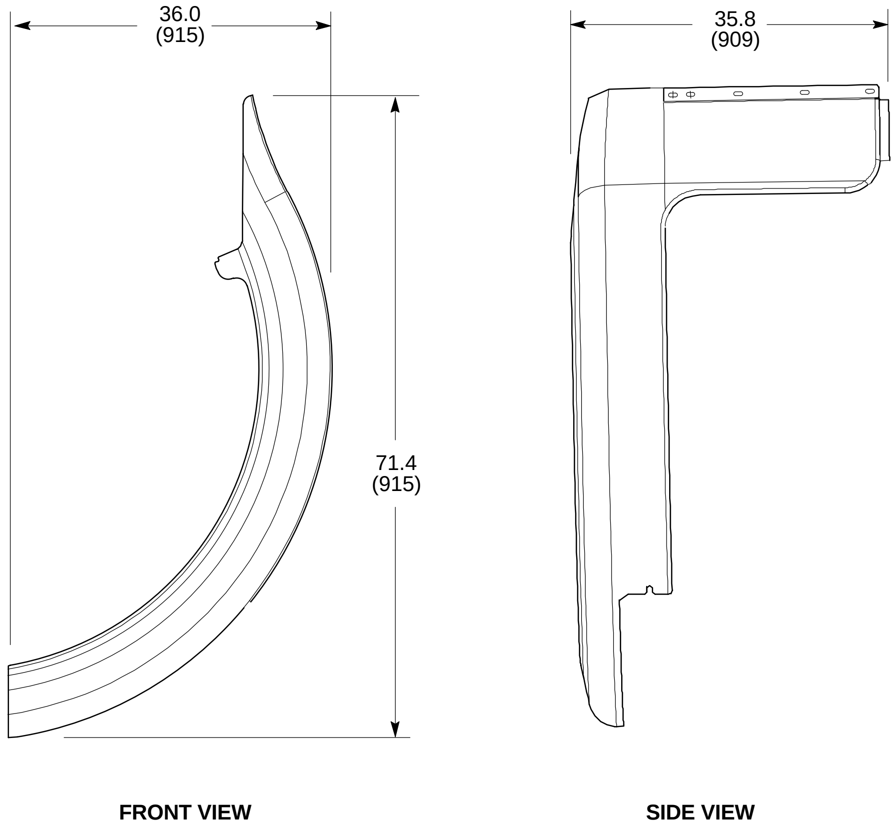
NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



SIDE VIEW **FRONT VIEW**

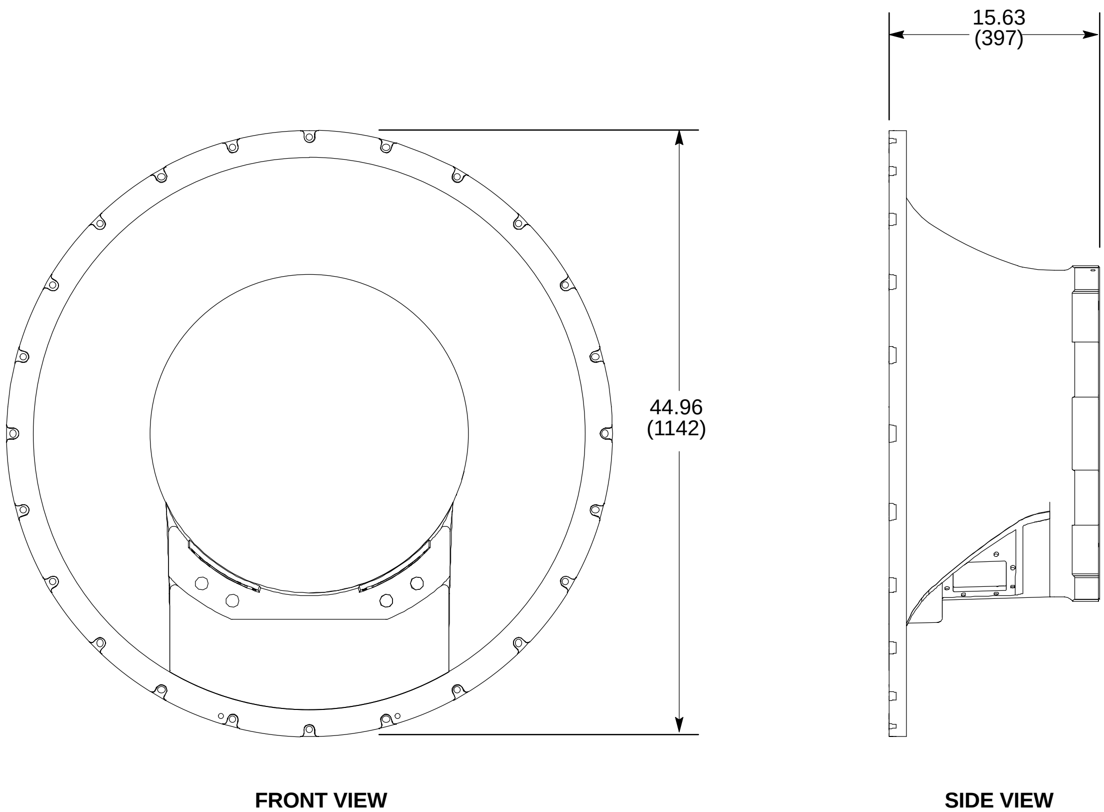
REAR RIGHT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2–25

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)

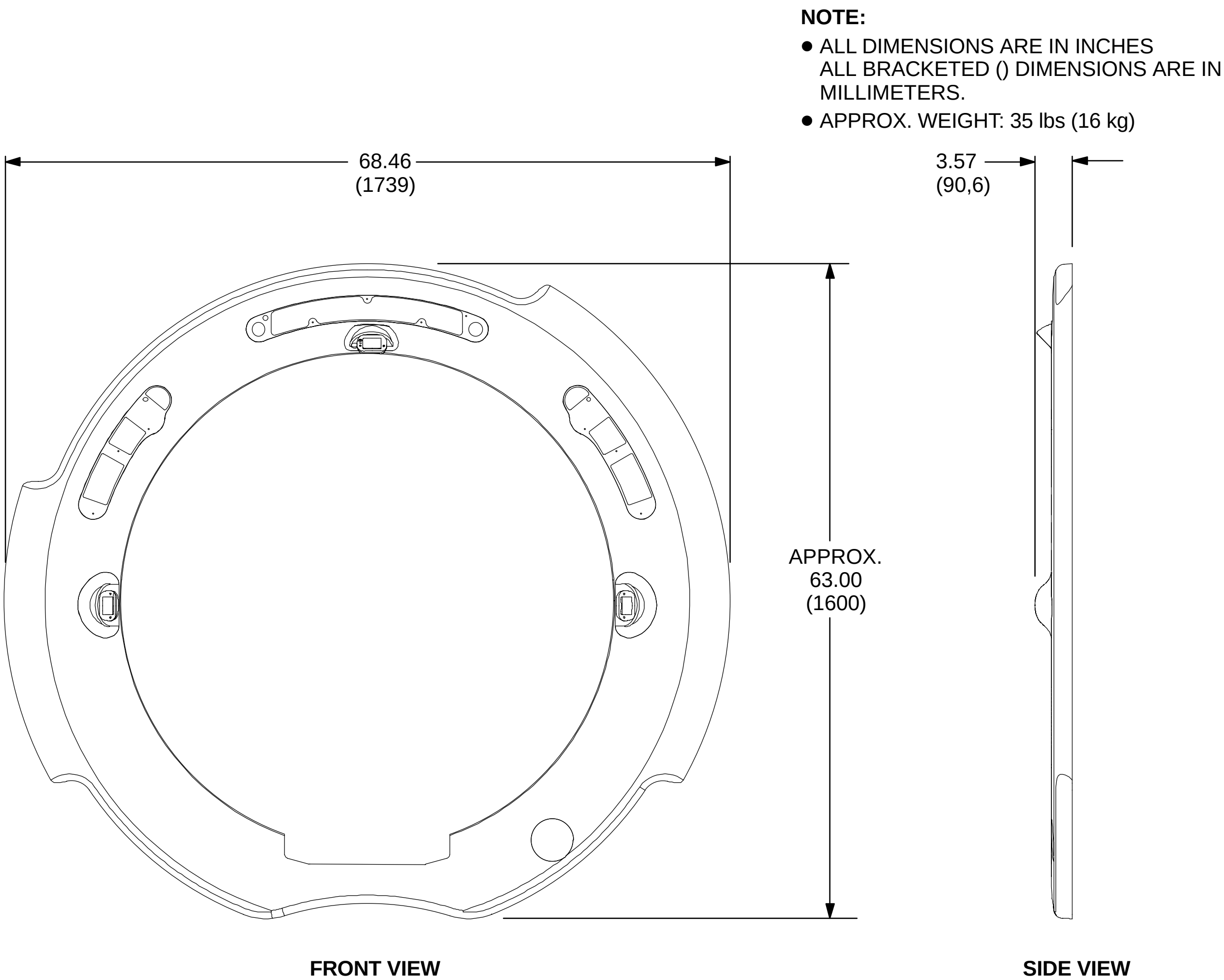


REAR LEFT ARC COVER – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2–26

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 70 lbs (32 kg)

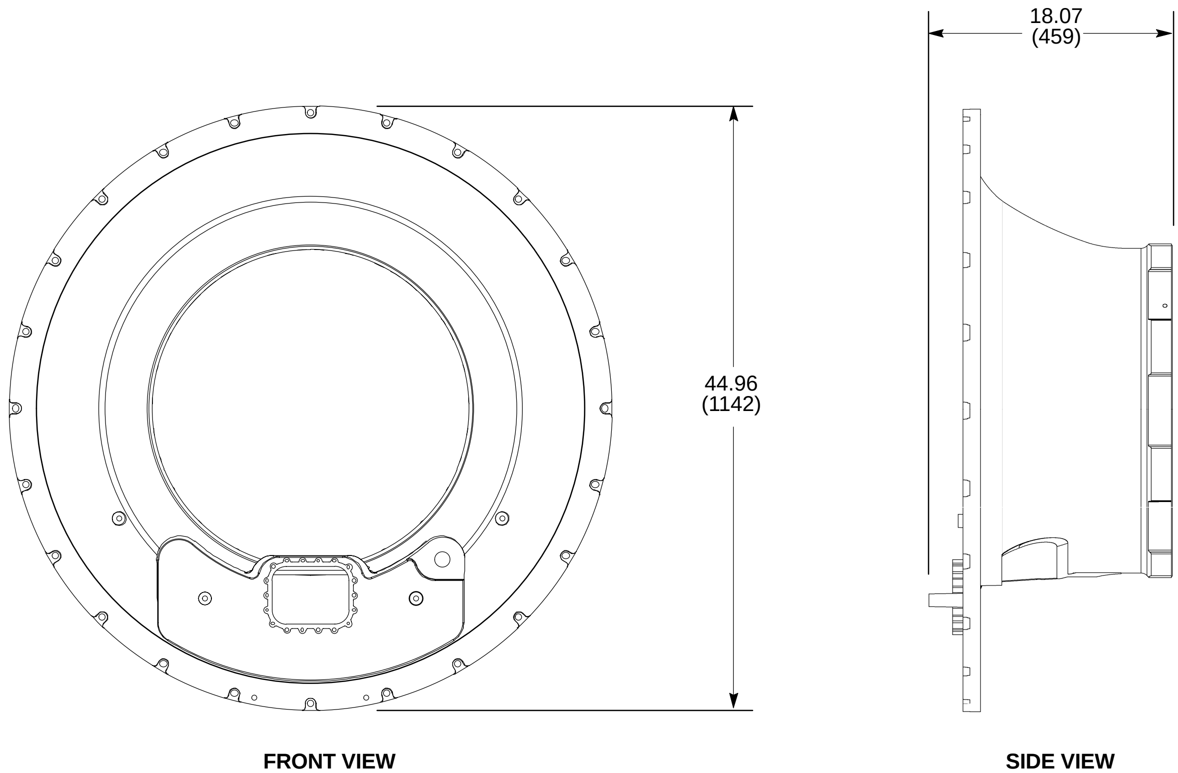


FRONT ENDBELL ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2–27



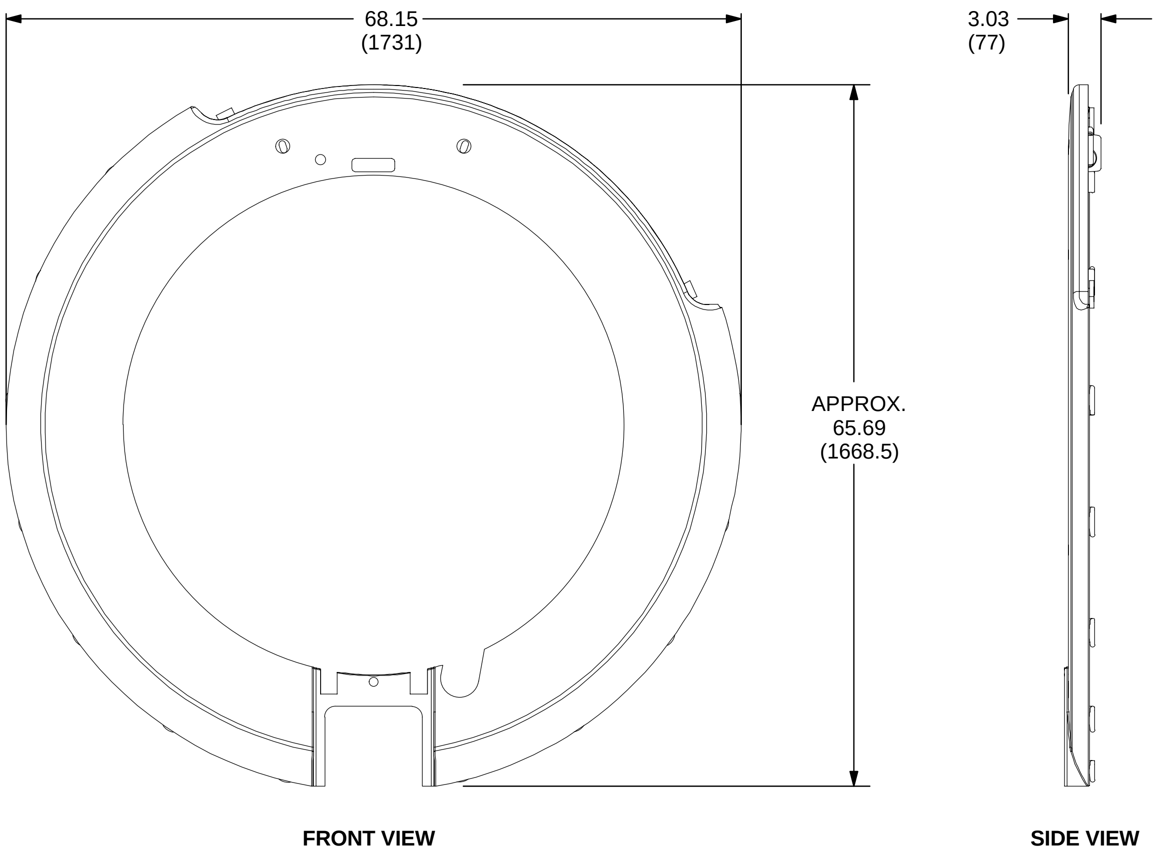
FRONT COSMETIC COVER ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2–28

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 75 lbs (34 kg)



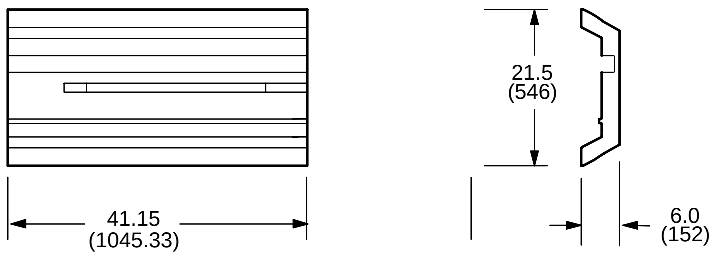
REAR ENDBELL ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2–29

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 35 lbs (16 kg)



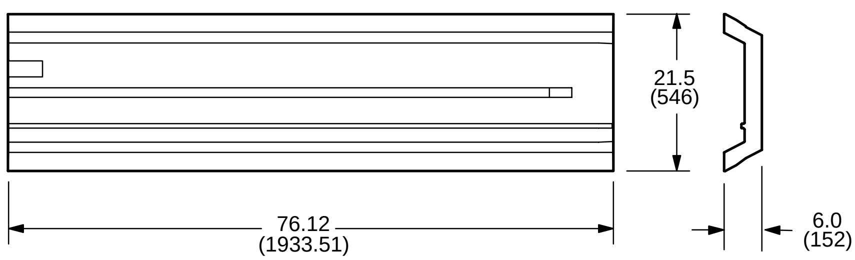
REAR COSMETIC COVER ASSEMBLY – LCC MAGNET QUIET TECHNOLOGY ENCLOSURE
ILLUSTRATION 2–30

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 75 lbs (34 kg)



TOP VIEW **END VIEW**

REAR PORTION SPLIT BRIDGE

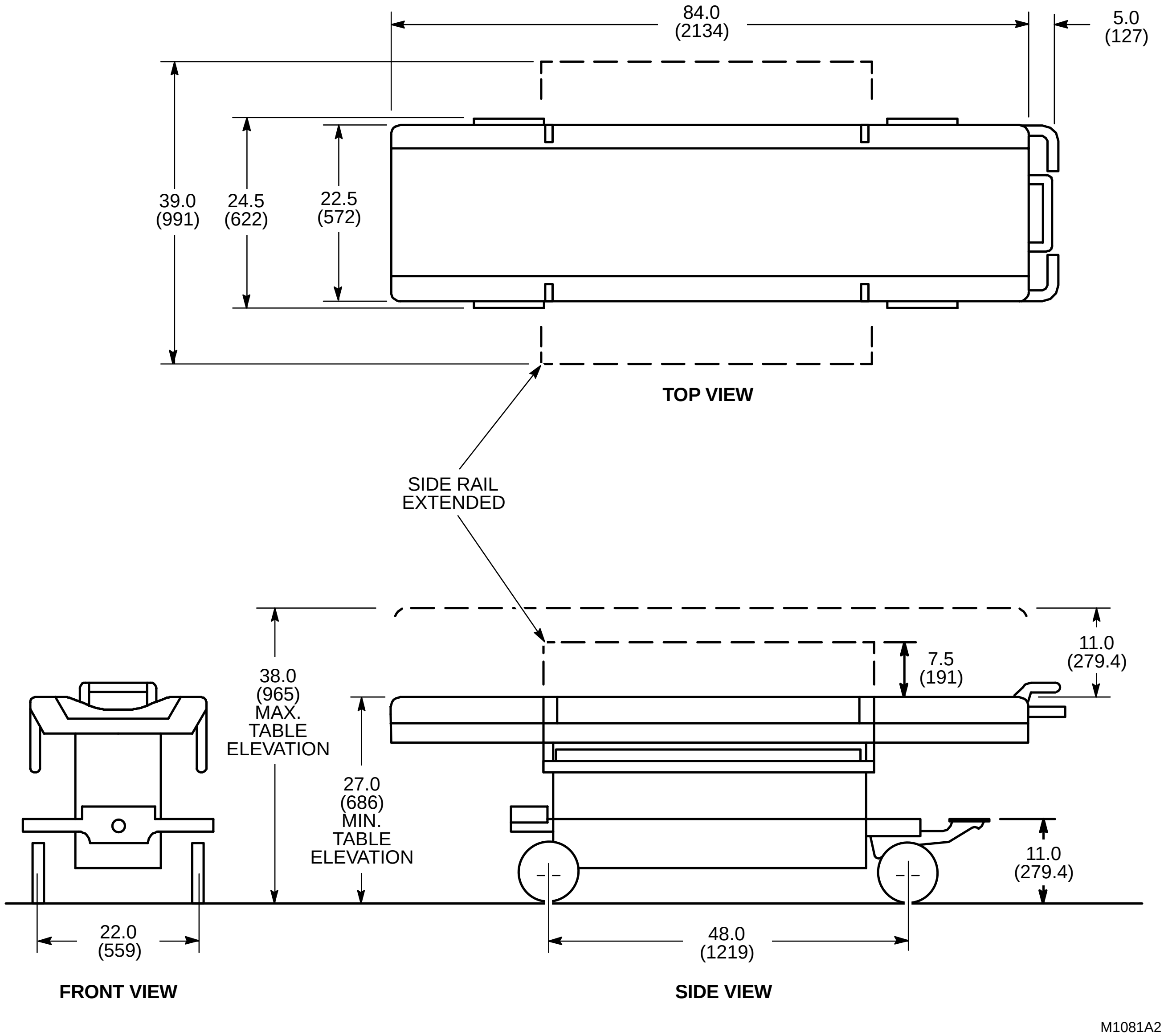


TOP VIEW **END VIEW**

FRONT PORTION SPLIT BRIDGE

SPLIT BRIDGE
ILLUSTRATION 2–31

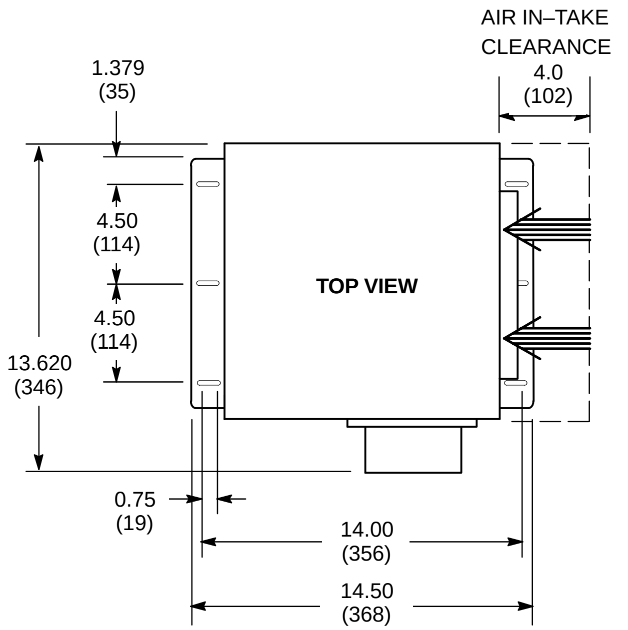
NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- WEIGHT: 280 lbs (127 kg)

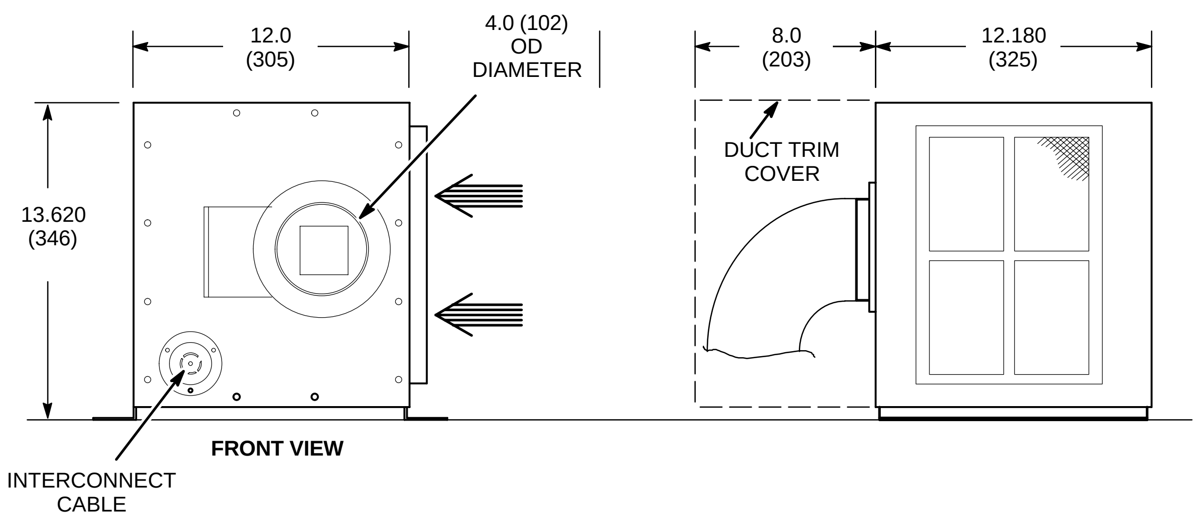


PATIENT TRANSPORT TABLE (PT1)
ILLUSTRATION 2–32

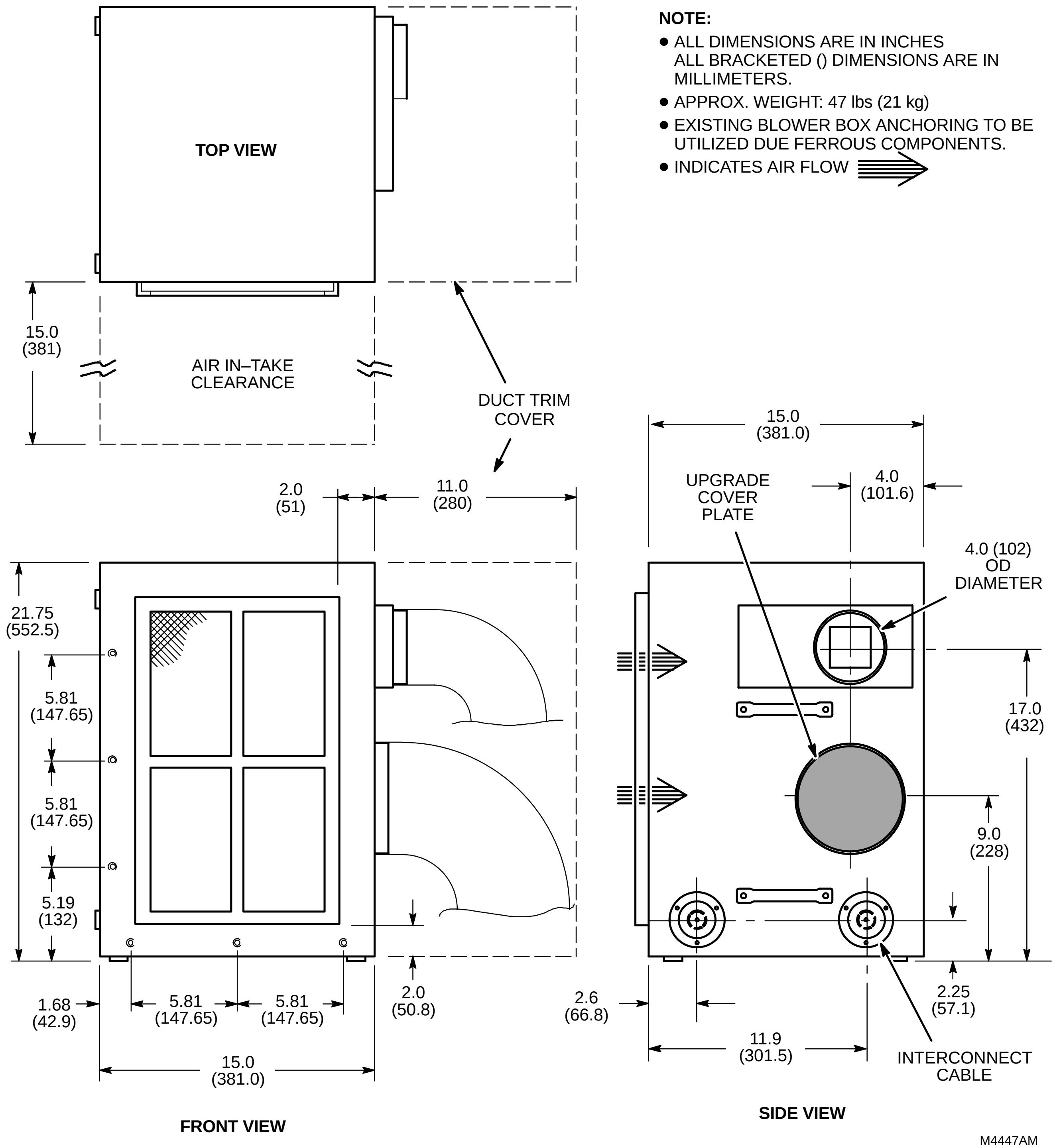
NOTE:

- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)
- BLOWER BOX MUST BE ANCHORED DUE FERROUS COMPONENTS.
- INDICATES AIR FLOW

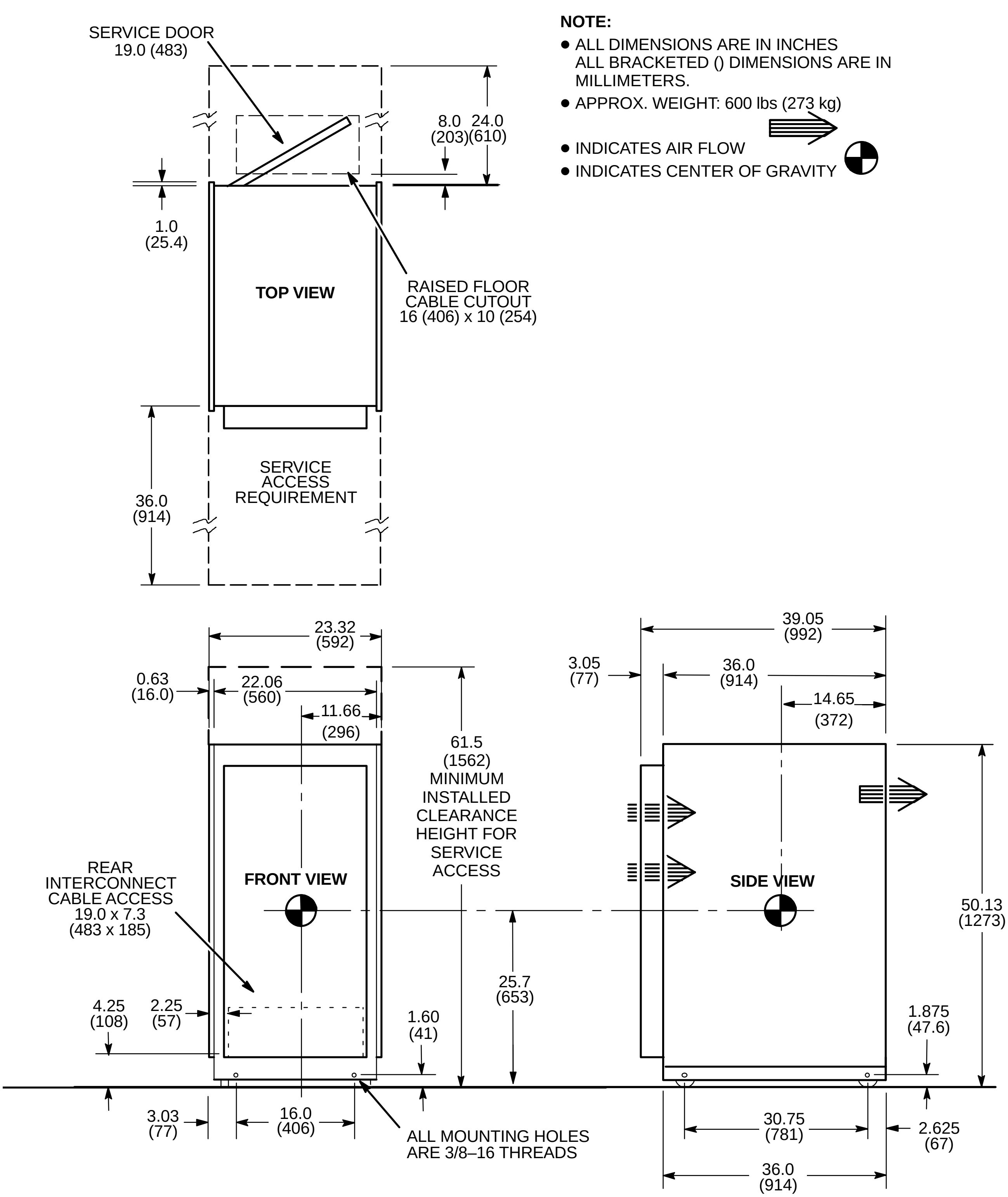




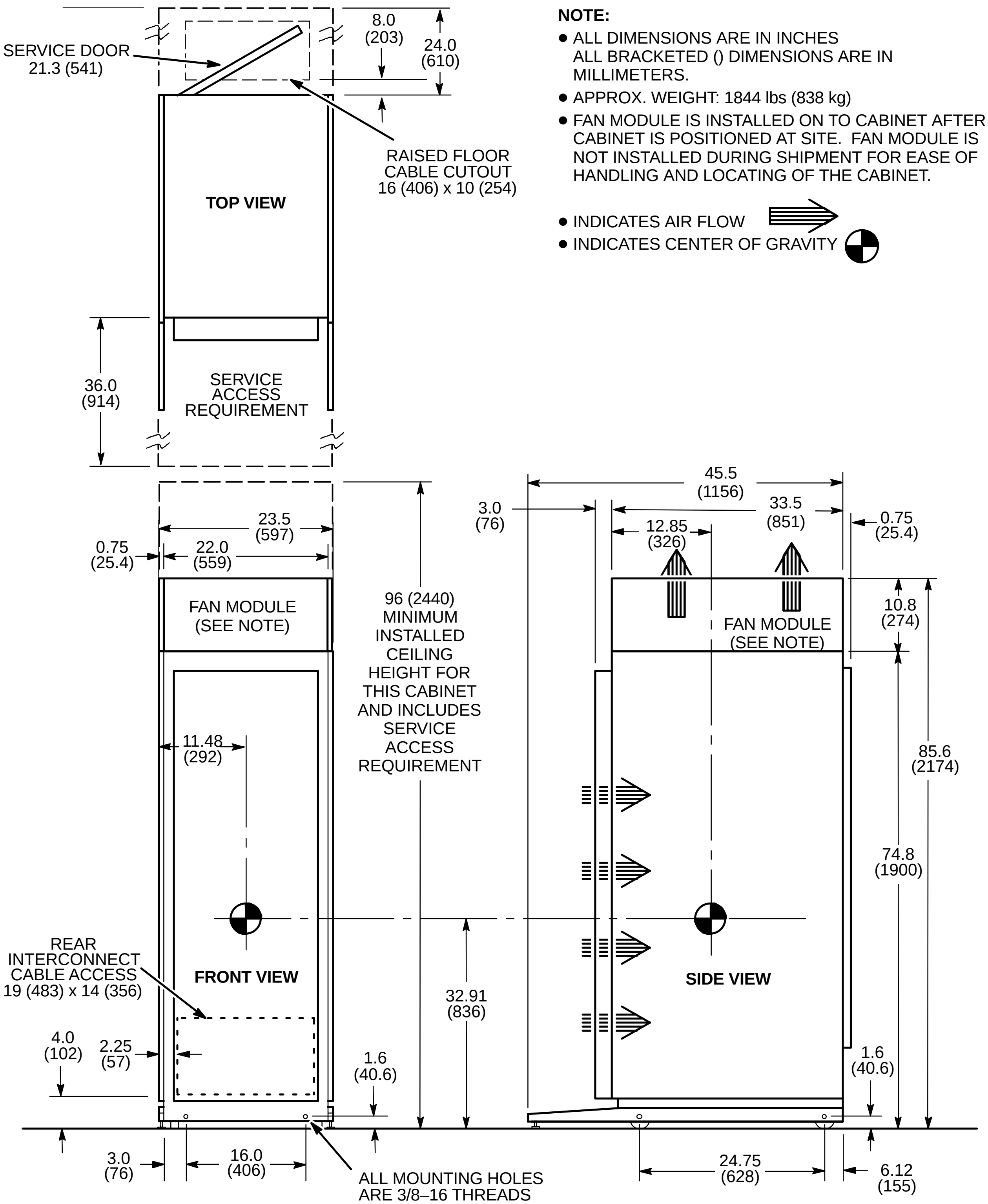
UPGRADE NEW BLOWER BOX (MG6)
ILLUSTRATION 2–33



UPGRADE MODIFIED BLOWER BOX (MG6)
ILLUSTRATION 2–34



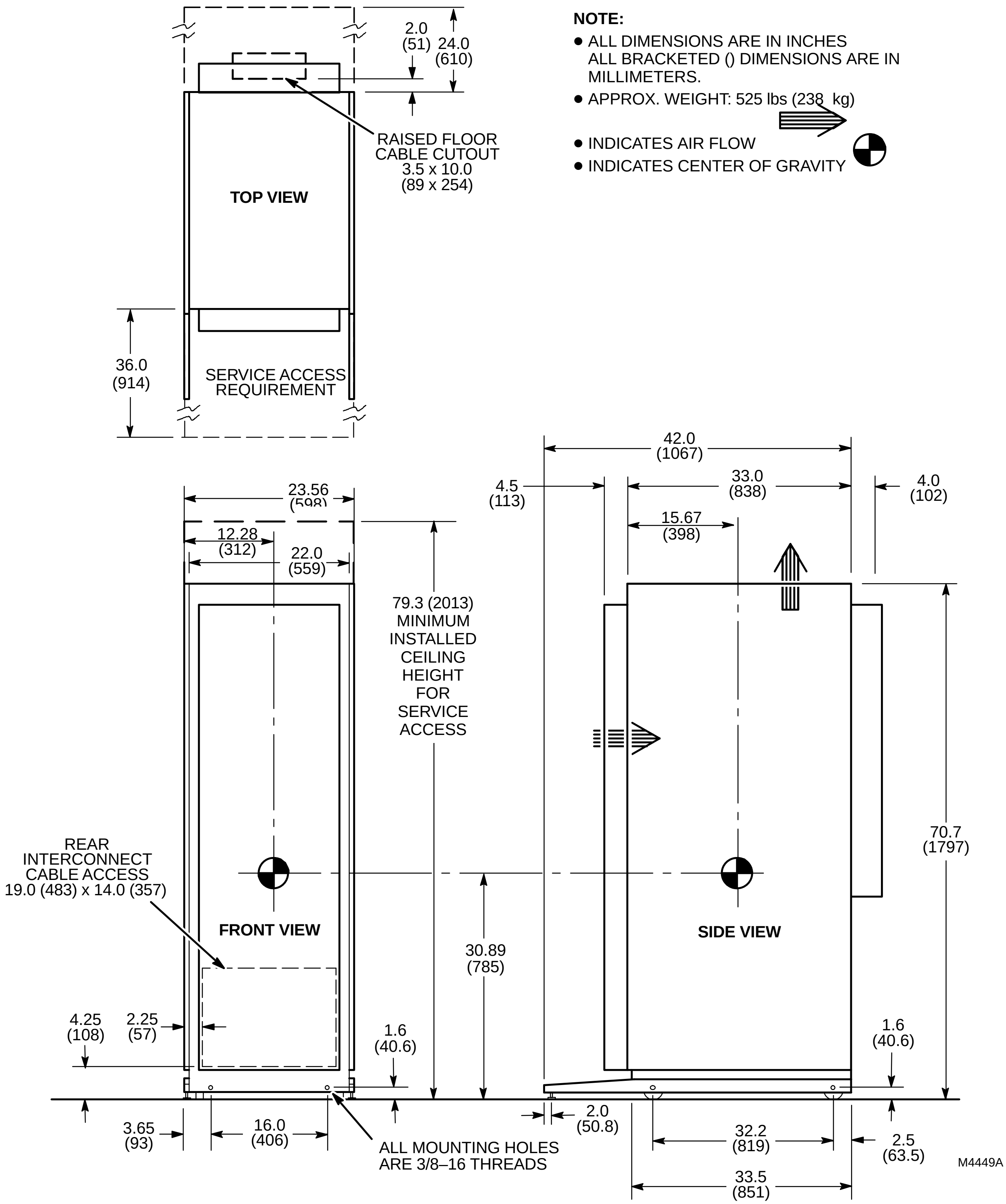
TWIN ACCESSORY CABINET (TAC)
ILLUSTRATION 2–35



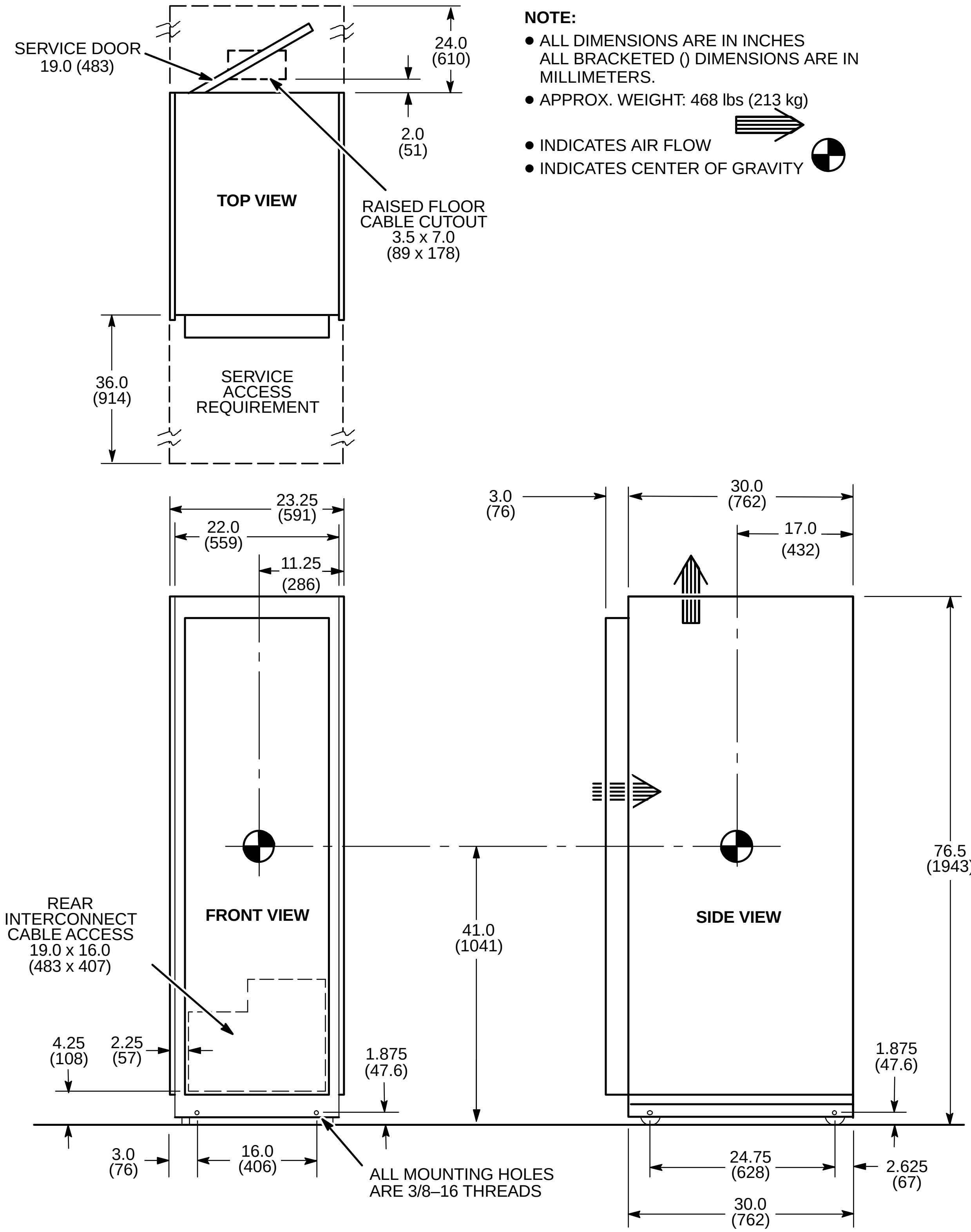
NOTE:

- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1844 lbs (838 kg)
- FAN MODULE IS INSTALLED ON TO CABINET AFTER CABINET IS POSITIONED AT SITE. FAN MODULE IS NOT INSTALLED DURING SHIPMENT FOR EASE OF HANDLING AND LOCATING OF THE CABINET.
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY

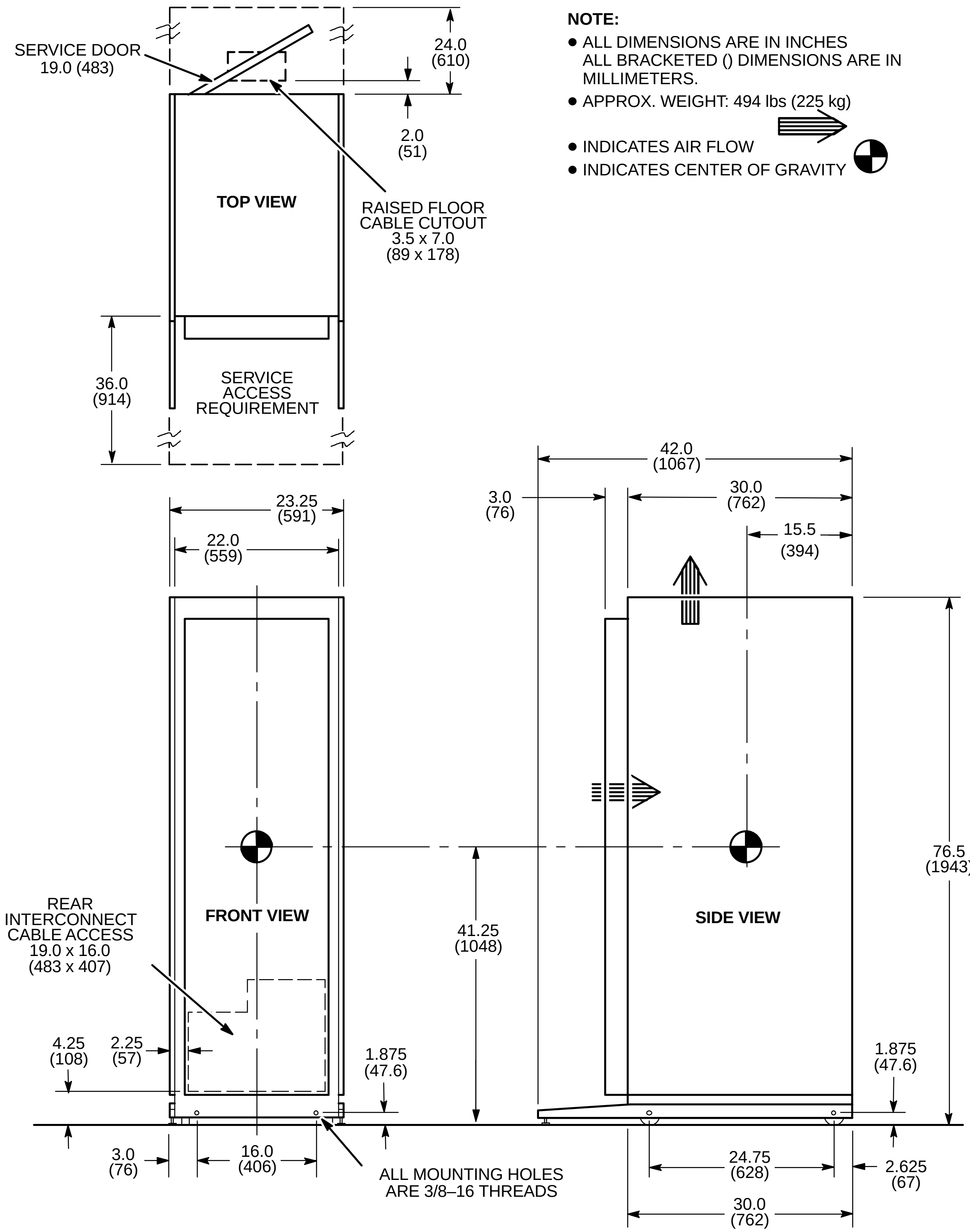
ACGD/PDU CABINET (MR3)
ILLUSTRATION 2–36



SRF CABINET (MR1)
ILLUSTRATION 2–37



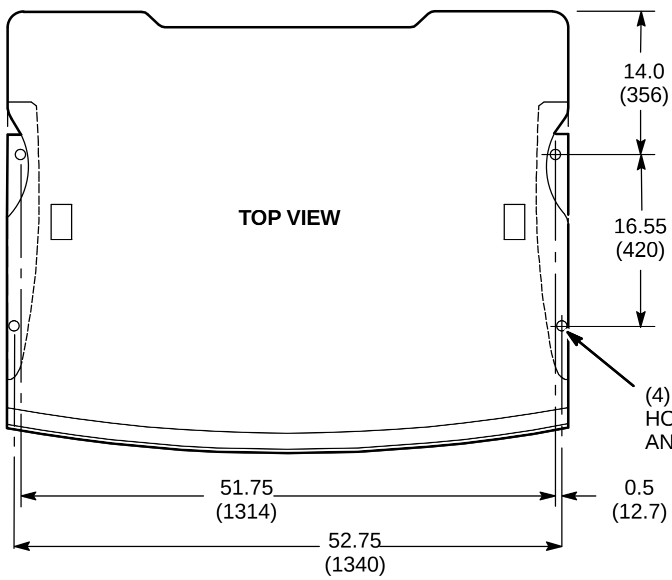
SYSTEM CABINET (MR2) FOR SIGNA TWIN*SPEED* WITH EXCITE TECHNOLOGY
ILLUSTRATION 2–38

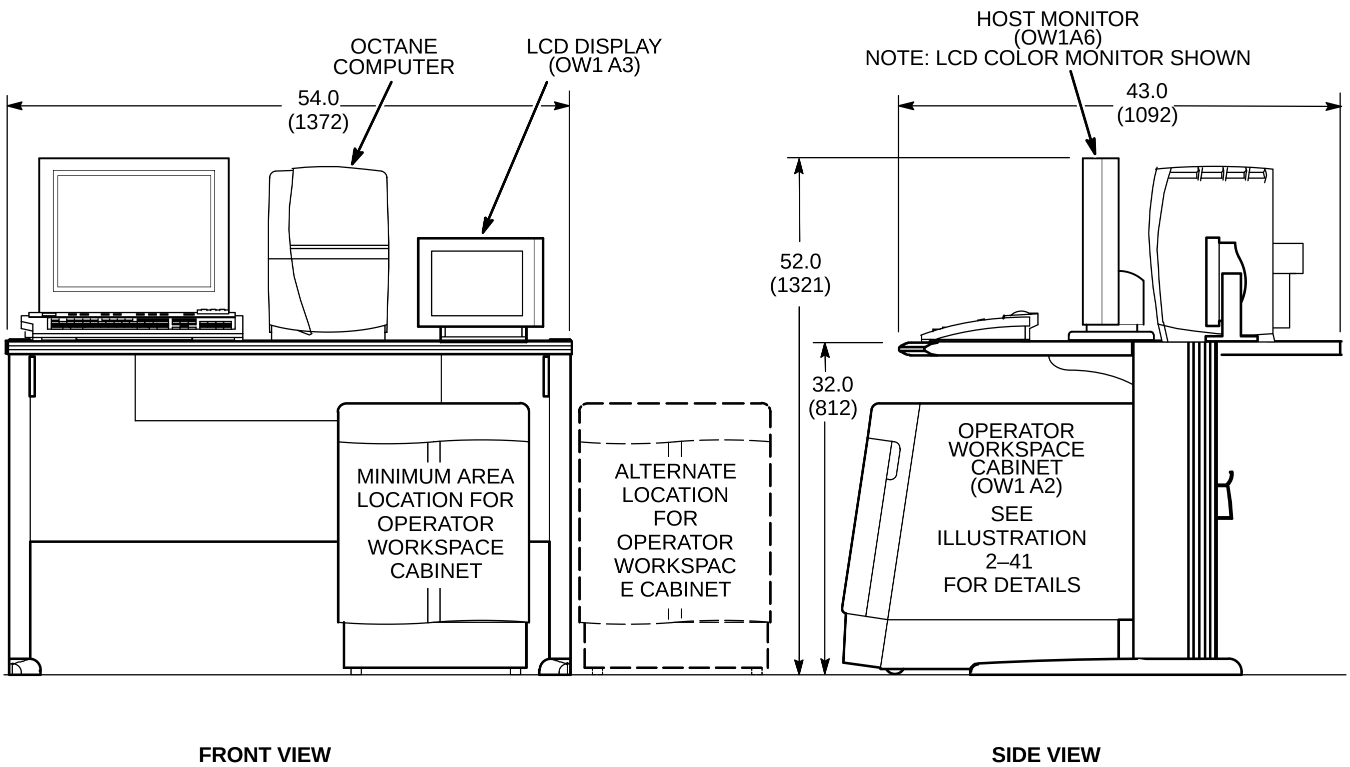


SYSTEM CABINET (MR2) FOR SIGNA TWIN*SPEED*
ILLUSTRATION 2–39

NOTE:

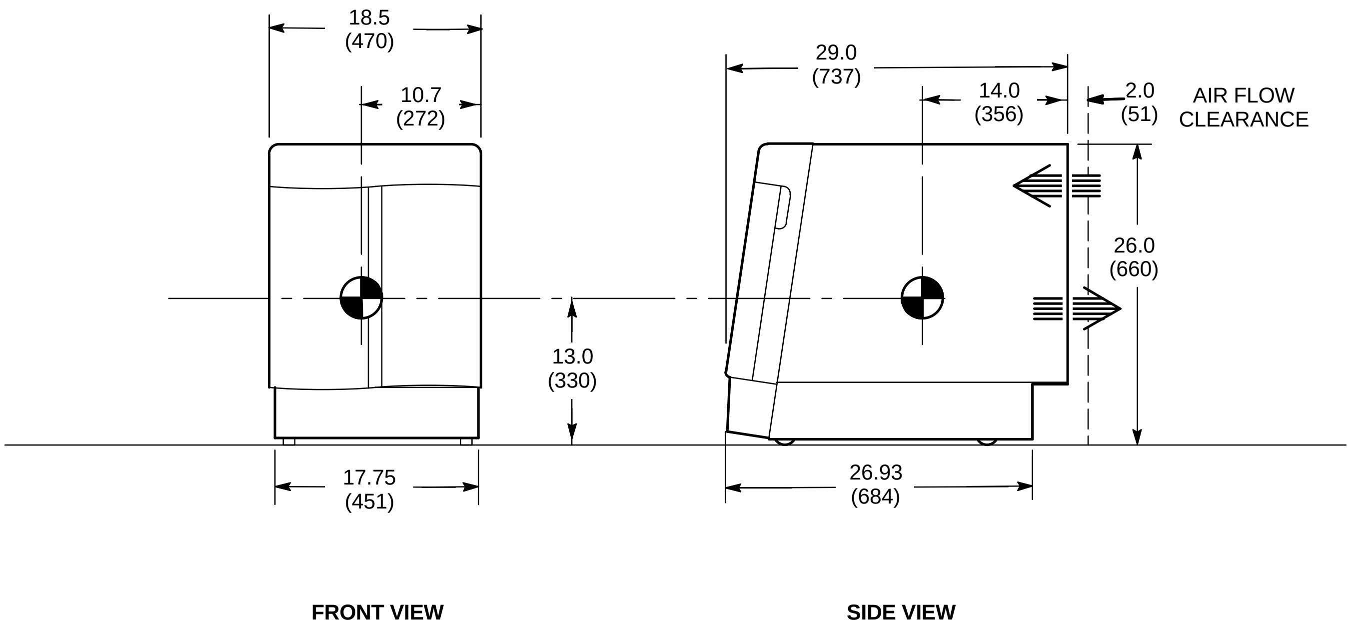
- ALL DIMENSIONS ARE IN INCHES.
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- ASSEMBLIES WHICH MOUNT TO UNDERSIDE OF TABLE AND IN TABLE CABLE TRAY ARE NOT SHOWN.
- OW TABLE IS AN INTEGRAL PART OF THE REGULATORY APPROVED SYSTEM. OW TABLE PROVIDES MOUNTING FOR SEVERAL ASSEMBLIES (E.G. OW INTERFACE MODULE, OW POWER DISTRIBUTION BOX, MODEM, DASM) AND CABLE ROUTING FOR OW INTERCONNECTS. DIMENSIONS OF EQUIPMENT LOCATED ON TOP OF OW TABLE ARE PROVIDED IN ILLUSTRATIONS 2–42 THROUGH 2–44 OW TABLE TOP LAYOUT PLANNING PURPOSES.



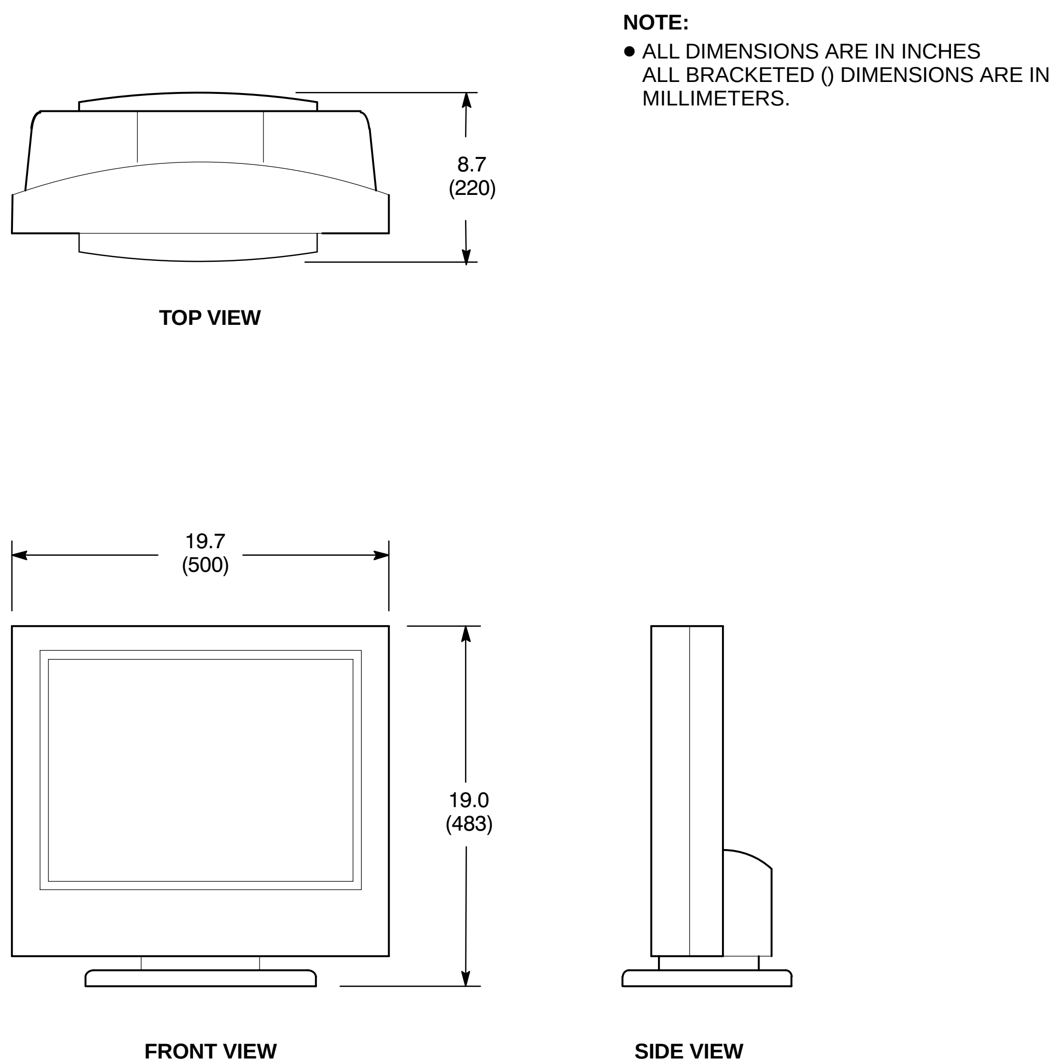


OPERATOR WORKSPACE (OW1)
ILLUSTRATION 2–40

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 192 lbs (87 kg)
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY

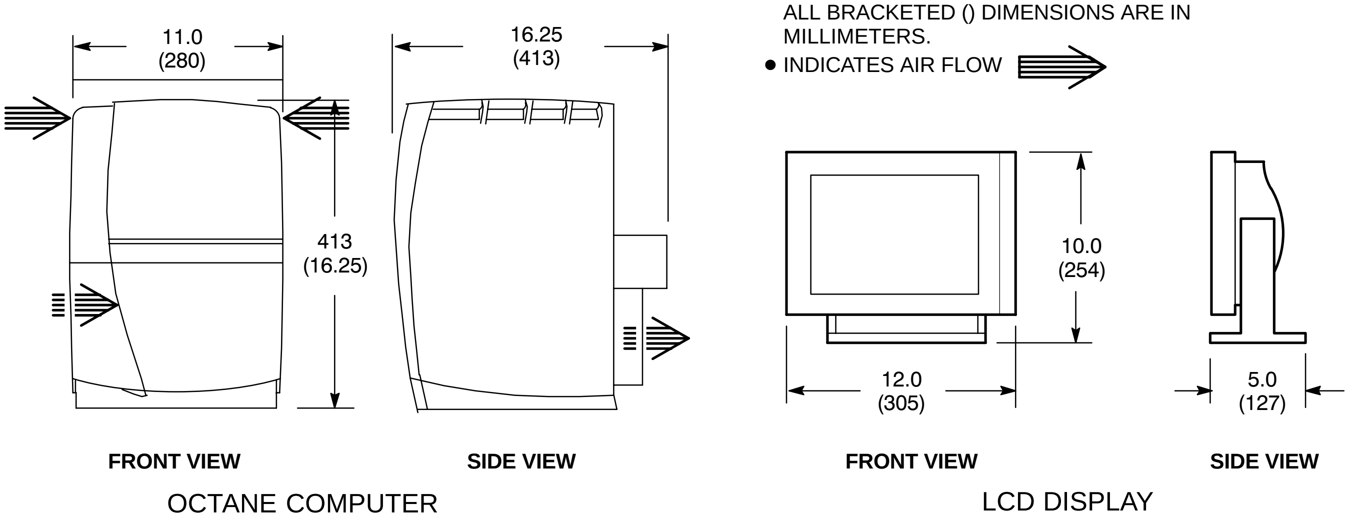


OPERATOR WORKSPACE CABINET (OW1 A2)
ILLUSTRATION 2–41



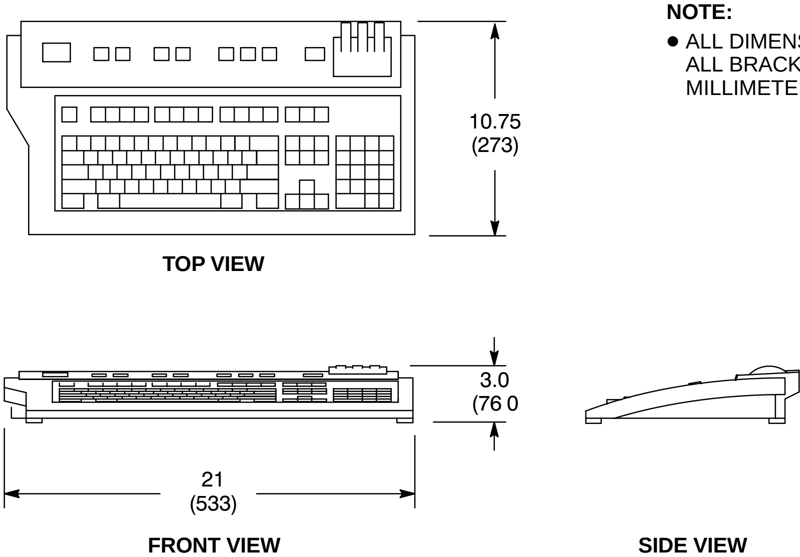
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – LCD COLOR MONITOR OPTION
ILLUSTRATION 2–42

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- INDICATES AIR FLOW

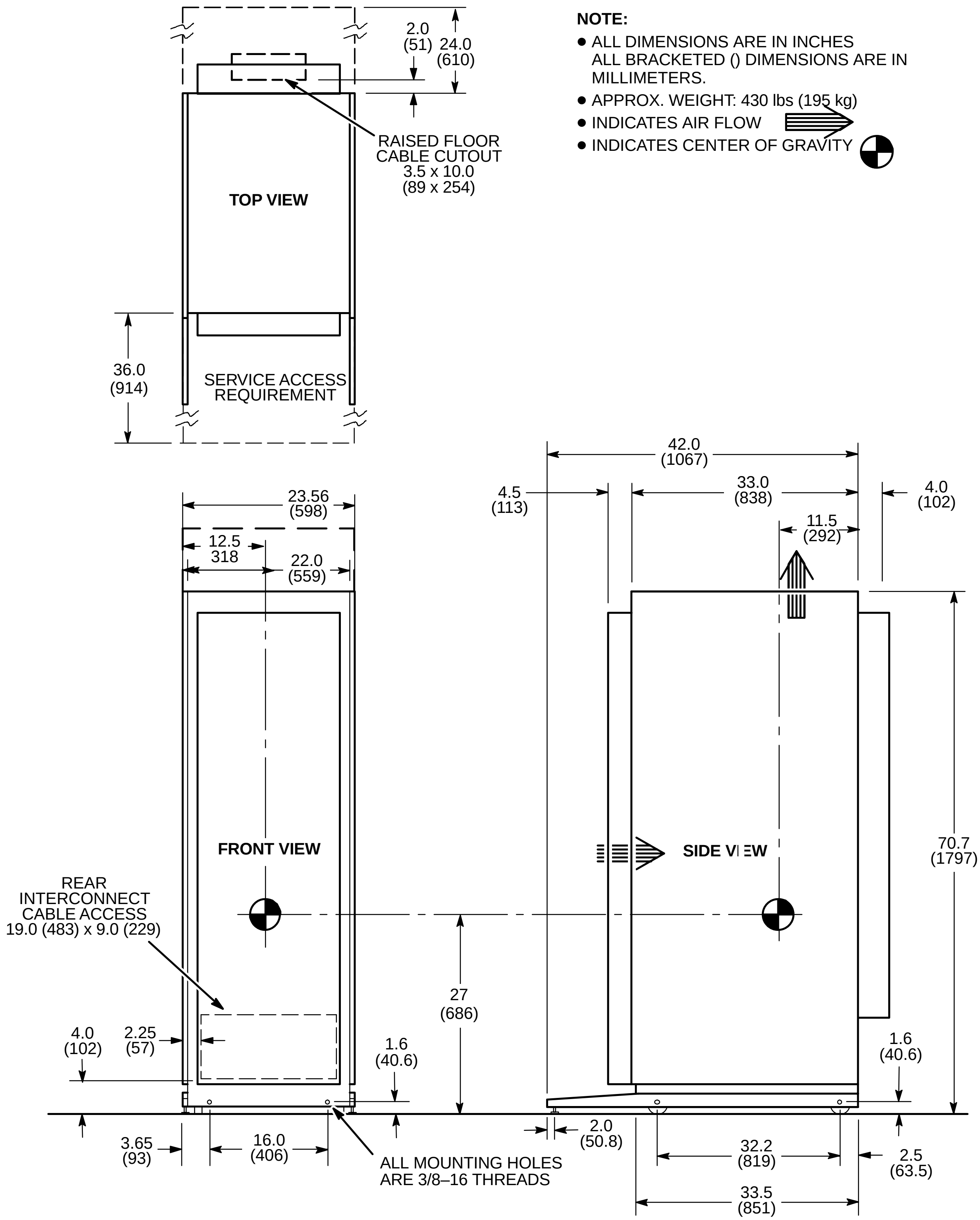


OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – OCTANE COMPUTER
ILLUSTRATION 2–43

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

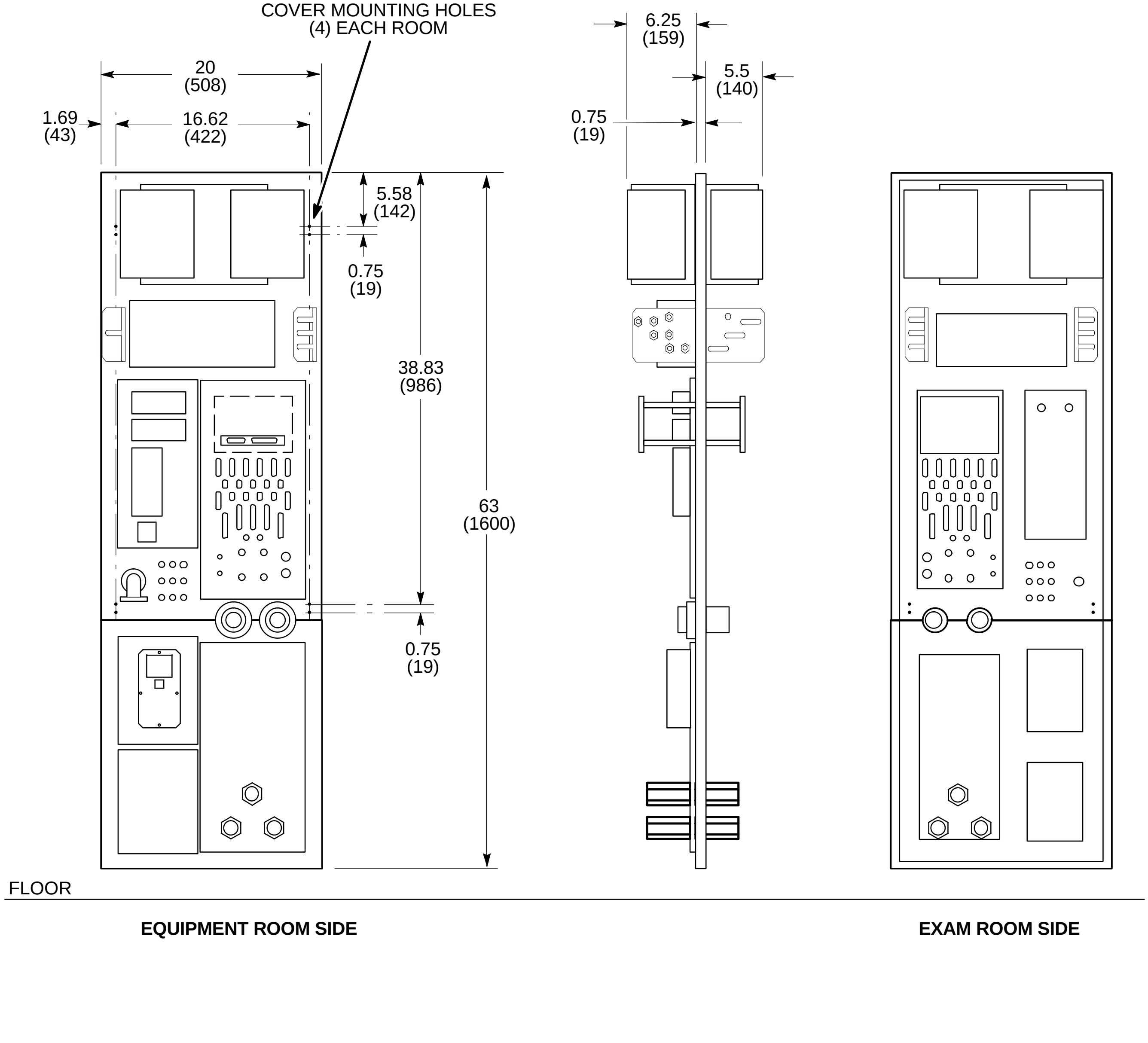


OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – KEYBOARD
ILLUSTRATION 2–44

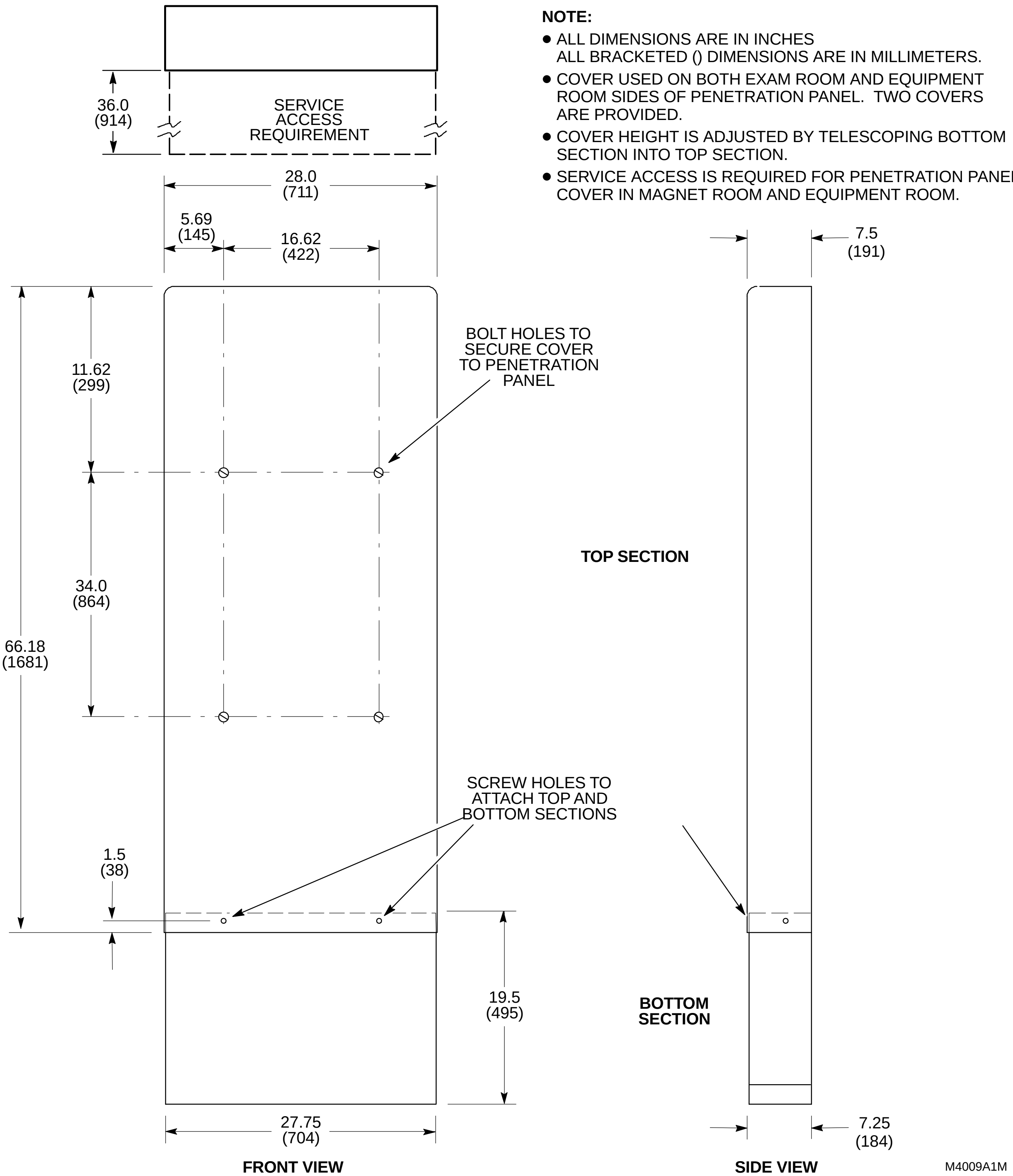


MNS CABINET (MNS) – SPECTROSCOPY OPTION
ILLUSTRATION 2–45

NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 52 lbs (23.6 kg)



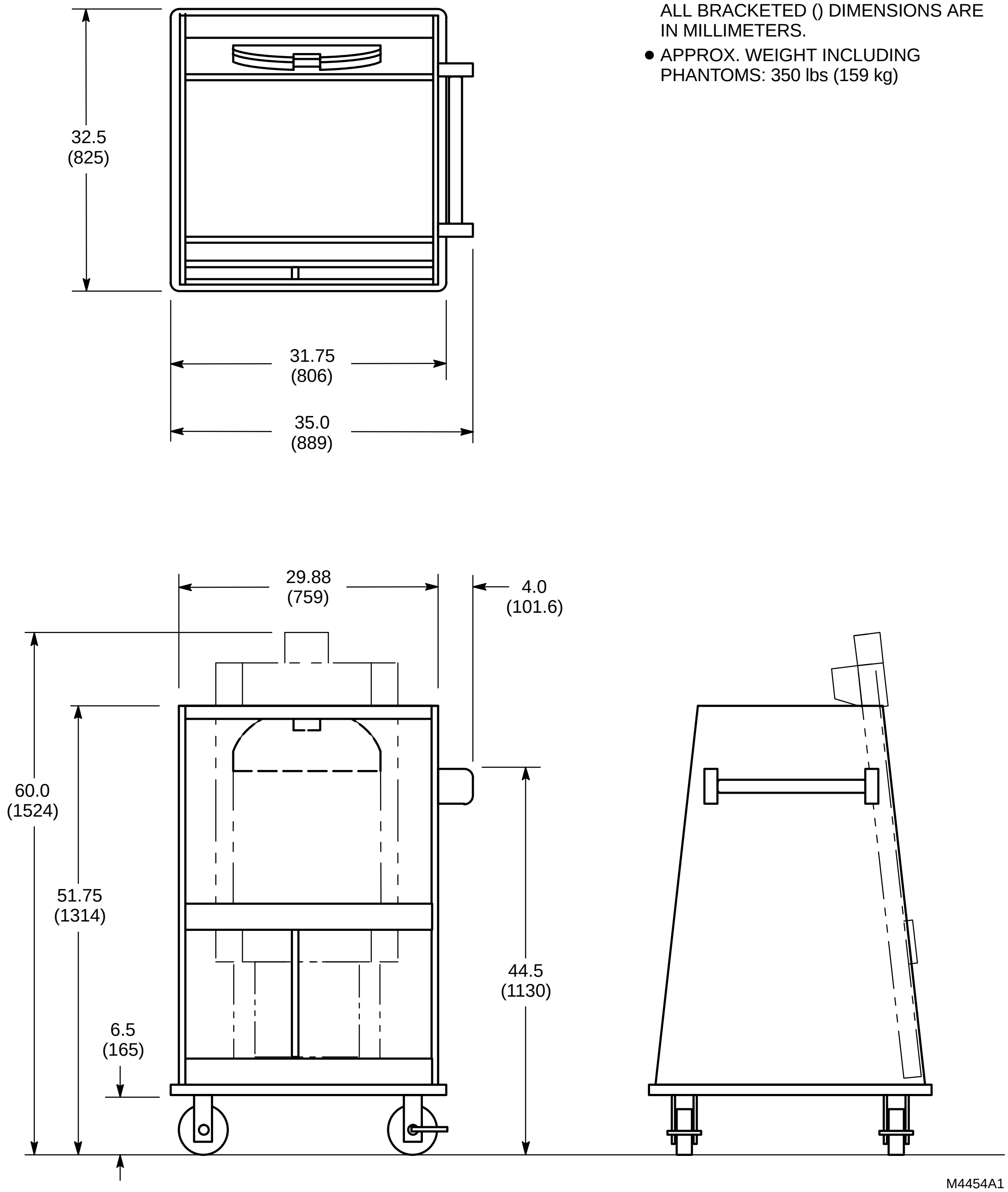
PENETRATION PANEL (PP1)
ILLUSTRATION 2–46



NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- COVER USED ON BOTH EXAM ROOM AND EQUIPMENT ROOM SIDES OF PENETRATION PANEL. TWO COVERS ARE PROVIDED.
- COVER HEIGHT IS ADJUSTED BY TELESCOPING BOTTOM SECTION INTO TOP SECTION.
- SERVICE ACCESS IS REQUIRED FOR PENETRATION PANEL COVER IN MAGNET ROOM AND EQUIPMENT ROOM.

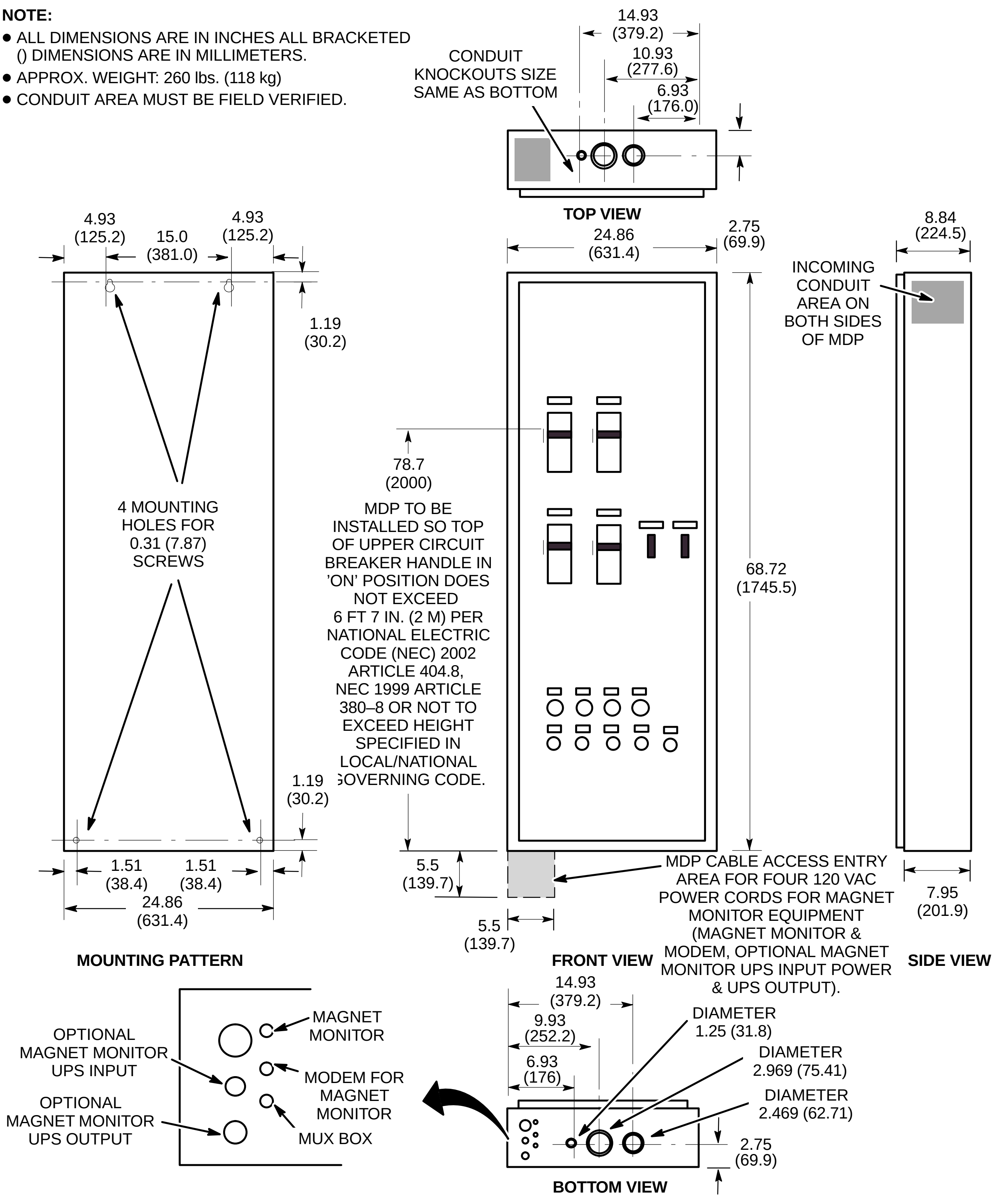
PENETRATION PANEL COVER
ILLUSTRATION 2–47

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT INCLUDING PHANTOMS: 350 lbs (159 kg)



SPT PHANTOM SET SHIPPING/STORAGE CABINET
ILLUSTRATION 2–48

NOTE:
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 260 lbs. (118 kg)
- CONDUIT AREA MUST BE FIELD VERIFIED.



SIGNA TWIN*SPEED* MAIN DISCONNECT PANEL (MDP)
ILLUSTRATION 2–49